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***Arab American University***

***Faculty of Engineering and Information Technology***

***Computer Systems Engineering***

***SENIOR PROJECT II***

***Soil Analysis and Monitoring System Using a Smart Vehicle***

***2026***

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***2026***

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I also, hereby that I have **read, reviewed and corrected the technical content** of this report and I believe that it is adequate in scope, quality and content and it is in alignment with the ABET requirements and the department guidelines.

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## **ABSTRACT**

The farmers and agricultural experts have always faced challenges in acquiring real-time soil information that can aid them in making the right decisions. Soil sample testing normally involves physical collection and lab testing, and thus time-consuming, costly, and labor-intensive.

In order to solve these problems, this work will develop a smart system for soil data acquisition and monitoring using a remotely controlled excavator vehicle. This vehicle will be fitted with an extendable robotic arm fitted with a multi-sensing system that acquires several soil parameters. These parameters include the soil moisture level, soil temperature, air Temperature and air Humidity.

In addition to this, the system will use a GPS system to determine the soil parameter reading location while the ESP32 micro-controller will process the acquired data and communicate it to a mobile phone app using Wi-Fi and Bluetooth communication systems. Besides, machine learning techniques will be utilized in the intelligent processing of soil data to facilitate smarter decisions.

This work will develop a practical, user-friendly, affordable smart system for application in smart agriculture.

### **Keywords:**

Soil analysis, Artificial Intelligence, Smart Vehicle, Soil Sensors, Mobile Application, Cloud Database, IOT(Internet Of Things).

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## LIST OF SYMBOLS AND ABBREVIATIONS

| LIST OF SYMBOLS |                                  |
|-----------------|----------------------------------|
| ₪               | Shekel                           |
| pH              | Measure of acidity or alkalinity |
| °C              | Degree Celsius                   |
| %               | Percentage                       |
| N               | Nitrogen                         |
| P               | Phosphorus                       |
| K               | Potassium                        |

| LIST OF ABBREVIATIONS |                                                   |
|-----------------------|---------------------------------------------------|
| AI                    | Artificial Intelligence                           |
| IEEE                  | Institute of Electrical and Electronics Engineers |
| MS                    | Microsoft                                         |
| IoT                   | Internet of Things                                |
| GPS                   | Global Positioning System                         |
| ML                    | Machine Learning                                  |
| Wi-Fi                 | Wireless Fidelity                                 |
| UI                    | User Interface                                    |
| UX                    | User Experience                                   |
| ESP32                 | Espressif 32-bit Microcontroller                  |
| NPK                   | Nitrogen, Phosphorus, and Potassium               |
| DHT22                 | Digital Temperature and Humidity Sensor           |
| DS18B20               | Digital Temperature Sensor                        |
| HW-080                | Soil Moisture Sensor Module                       |
| RC                    | Remote Control                                    |
| CSV                   | Comma-Separated Values                            |
| CPU                   | Central Processing Unit                           |
| DC                    | Direct Current                                    |

# 1 CHAPTER 1: INTRODUCTION

Modern technology has been widely used in the development of the agricultural sector in various ways. It concerns mostly soil analysis and soil monitoring that have been improved by using some of the latest developments in technology[1]. The process of conventional soil tests includes manually collected samples sent to laboratories, and it can be considered relatively slow and expensive procedure compared to soil monitoring technologies[2]. That is why the introduction of technologies plays the key role in providing more accurate and quick data about soils.

One of the biggest advantages of smart soil monitoring lies in the opportunity to collect data right from the field that allows making better conclusions regarding soil characteristics. Besides, technologies like wireless communication and intelligent monitoring can be used to improve the process of collecting the information and analyzing it later. The combination of those tools makes the analysis much quicker and easier[1].

The suggested device consists of a remote-controlled excavator machine with a movable arm attached to it. A special multi-parameter soil sensor is attached to that arm, and the information received from the sensors is sent through the air to a mobile application installed on the farmer's smartphone.

## 1.1 Problem Statement and Purpose

Soil data constitute one important part of crop assessment and decision making process. As far as soil data gathering goes, the conventional method of gathering such data mainly involves soil sampling and it is quite cumbersome as well as costly process. In addition, the delays involved in processing the gathered information about soil samples can contribute significantly to lowering down the accuracy of such data[2].

The goals for the proposed project include designing and developing an intelligent system of soil monitoring and analysis using a smart mobile platform. Such a system is supposed to collect soil data in situ and in real-time by integrating sensor technology, wireless

communication technology, and GPS[1]. It can be predicted that through using the automation method for collecting soil data, as well as direct access to this data via apps on mobile phones, this suggested project would increase efficiency in soil analysis in agriculture.

### 1.2 Project and Design Objectives

The main goal and idea of this project is to create an intelligent and integrated system for performing soil analysis and monitoring using a mobile vehicle, taking into consideration various engineering and agricultural demands. The main focus of this project is to apply engineering, Internet of Things, and smart technologies for performing soil analysis[1].

In order to accomplish all of the above, the design and functional objectives of the project would be:

- Utilization of a ready-made remote-controlled excavator vehicle to reach various agricultural areas and make soil measurements there.

Incorporation of soil sensors for measuring soil moisture, soil temperature and air sensor to measuring air temperature, and air humidity.

- Creation of control software for collecting, processing and transmitting data from sensors on the basis of ESP32 micro-controller.
- Providing for wireless communication between control system and the application through Wi-Fi and Bluetooth technologies.
- Linking each measurement with geographic location using GPS module.
- Performing soil analysis using machine learning techniques.
- Designing of easy-to-use mobile application that will visualize soil readings and results of analysis.
- Ensuring possibility to expand system with further additions in the future.

### 1.3 Intended Outcomes and Deliverables

At the end of the project, the following products will have been produced:

- A working prototype of a mobile vehicle for soil sampling with the integration of sensing and communication systems.
- Cross-platform mobile application with real-time data visualization, storage, and AI-based predictive analysis.
- A database or database system that is cloud based, storing soil data that can.
- Complete technical documentation, source code, CAD drawings, schematics, and a comprehensive user manual.
- Field demonstration of its ability to perform well in difficult terrain and to yield results considerably more quickly than conventional laboratory-based approaches.

### 1.4 Summary of Report Structure

The report consists of seven chapters and each chapter covers a specific aspect of the project for a soil analysis and monitoring system using a smart vehicle.

Chapter One provides an overview of the motivation or reason for undertaking this project, the nature of the problem being studied, the objectives, and the expected outcome, thus providing the necessary background information or general context for providing a solution.

Chapter Two highlights the existing technique for soil testing/monitoring with respect to agricultural purposes. In addition, past research and studies will be highlighted concerning soil testing/monitoring, along with a comparison of existing similar smart systems that exist with regard to the proposed system.

, Chapter Three explains the technical methodology used in the development of the system. It explains the system architecture, hardware components, software components, system design

specifications, standards used, design constraints, design alternatives, and finally, the basis for selecting the best design through engineering analysis and feasibility studies.

Chapter Four features the results obtained from implementing and testing the system, which include results concerning sensors, data transmission, and the proposed soil analysis technique. Besides, there is a discussion on how well such a system would perform in relation to other approaches.

Chapter Five highlights the role of project management, which includes task planning, scheduling, and costing, together with lessons learned in the project development stages.

Chapter six discusses in detail the general implications of the proposed engineering solution in terms of economy, society, nature, and ethics, with emphasis on the system's contribution to improving agricultural production, supporting engineering decisions, and minimizing human effort and resources.

## 2 CHAPTER 2: BACKGROUND

This chapter is necessary and useful background material to assist the reader's understanding of the basic concepts upon which the research of the Smart Vehicle Soil Analysis and Monitoring System is based. Additionally, relevant existing systems and scientific research in the concept of smart vehicle soil analysis and monitoring will be fully discussed. Section 2.1 of this chapter will deal with the essential theoretical and technological concepts that can aid the understanding of the research on the smart vehicle soil analysis and monitoring system. Section 2.2 of this chapter will address the relevant existing systems and previous scientific research existing on smart vehicle soil analysis and monitoring. Furthermore, a comparison of existing relevant research will be fully discussed with regards to the scientific contribution of the proposed research.

### 2.1 Overview

To ensure high-quality soil samples and, subsequently, improved agricultural productivity, it is crucial to accurately monitor the environment using various technologies in precision agriculture and smart farming applications. While monitoring soil conditions, it is important to measure specific parameters, for example, soil moisture, temperature, air moisture and temperature. Additionally, it is crucial to monitor energy consumption due to high energy demand from various sensors, actuators, and data transmission.

This project intends to design an effective and reliable system for monitoring the environment with the integration of different sensors, including soil moisture and temperature sensors, air moisture and temperature to monitor the soil conditions persistently, wherein the data collected will be sent wirelessly for further processing and analysis.

The system can be used through a mobile application, which will allow users to monitor their data and system functionality. The proposed system is useful to farmers and agricultural engineers, enabling them to monitor the data and make swift decisions. There are also some advantages to using this system, such as collecting soil sample data with the help of a mobile vehicle, enhancing agricultural production, and affecting agricultural and engineering projects.

## 2.2 Related Work

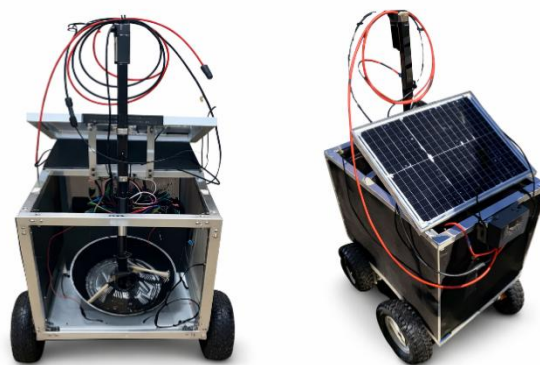
With the development of smart sensing technology, IoT, and robot technologies, various technologies have emerged to assist in the analysis of the soil, especially in agriculture and engineering. This chapter provides an overview of current systems and related research studies, analyzing their scope, limitations, and methodologies within the context of smart soil analysis and monitoring systems.

### 2.2.1 Soil Sampling and Analysis Robot S.S.A.R

They designed a robot to collect these soil samples and analyze them without the need for human hand. it is used in any field that requires analysis of soil conditions such as in agricultural, construction, and environmental areas.

The principle of this robot, first thing is drilling then inserts probes to the samples that we collected before from different depths, then analyzes the soil directly on site that combine with these following sensors such as: moisture, pH level, salinity, nitrogen, phosphorus, and potassium. The robot collect and stores these samples and sends them to the laboratory later. Based on the test results it will show us the soil properties at each point of the land.

If we have areas that are difficult to reach we can use artificial intelligence to help us in analyze soils [\[3\]](#).



*Figure 2.1 :S.S.A.R Front and Back View*

### 2.2.2 A Mobile Laboratory Robot For Various And Precise Measurements of Crops and Soil in Agricultural Fields :Development and Pilot Study.

Noda and colleagues (2023) developed a robot that is very useful for the protection from sunlight and wind to ensure measurements accuracy. Basically, this robot is capable of measuring soil electrical conductivity and capturing images of crops using RGB cameras. This robot has a modular sensor unit that can be swapped according to needs. For example, during a field trial in a cabbage field, the robot was able to collect data from different points and generate precise maps showing soil properties and crop conditions. These studies help in reducing manual labor, increasing accuracy and covering a large area of the field. Other studies focused on the significance of incorporating precise position system technologies coupled with the use of multi-sensor units to improve the flexibility and convenience of crop and soil analysis, creating a solid basis for the successful innovation of effective mobile robots [4].



*Figure 2.2 :Hardware configuration of a mobile laboratory robot [5]*

### 2.2.3 Path to autonomous soil sampling and analysis by ground robots.

This study focusses on the development of ground-based robots for collecting and analyzing soil samples without direct human intervention. This system is very accurate and reliable even in dangerous and hard-to-reach areas. Also, this research presents a comprehensive design methodology for these robots, based on four important components: sensing to detect contaminants, sampling to collect and prepare soil, mobility to navigate various terrains, and autonomy to determine optimal sampling locations and plan movement paths. This study can be

used to design tools for soil extraction and measurement for different soil types and sampling depths, along with selecting suitable mobility systems to ensure stability and prevent problems that maybe occur during navigation. These robots can highly measure soil contaminants, such as chloride, with results comparable to laboratory analyses. So basically this research enhance efficiency, reduce human risk, allow deployment in divers environments according to their flexible and adaptable design [6].

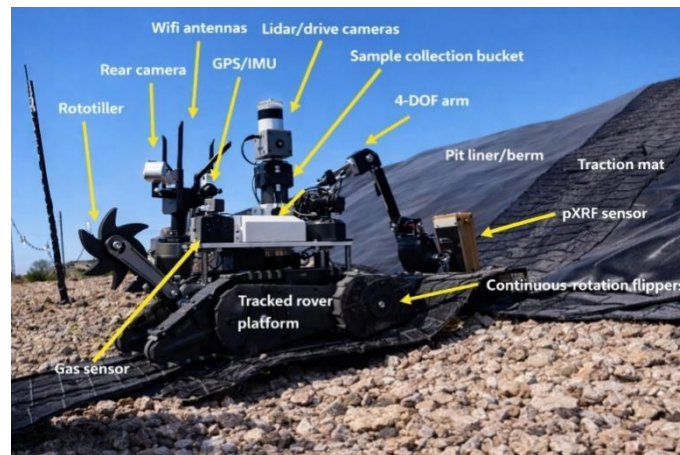


Figure 2.3 :Path to autonomous robot[6]

### 2.2.4 Kriging-Based Robotic Exploration for Soil Moisture Mapping Using a Cosmic-Ray Sensor.

The purpose of this research is based on a mobile robot equipped with a non-contact soil moisture sensor( of the "cosmic-ray neutron sensor" type) to measure soil moisture in the field without the need for thousands of manual samples or laboratory analysis.

A robot builds a "humidity map" of the site as it moves, automatically selecting the best points for additional measurement/data collection, based on the accuracy and inaccuracy of the current model [7].



Figure 2.4 :Kriging-Based Robotic [8]

Table 1 :Related Work Comparison

| Feature | S.S.A.R.                                                                               | A Mobile Laboratory Robot.                                       | Path to autonomous .                                                                                                                      | Kriging-Based Robotic.                                           | Our System                                                                                                                    |
|---------|----------------------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Purpose | Design a low-cost robot that collects soil samples and analyzes NPK nutrients on-site. | Automate and improve the accuracy of soil property measurements. | Develop an autonomous ground robot capable of performing soil sampling and on-site analysis to improve accuracy and reduce manual effort. | Autonomous robot that generates high-quality soil moisture maps. | To build a smart soil-inspection vehicle that collects soil data and supports both agricultural and engineering applications. |

|               |                                                                              |                                                                              |                                                                                                                                     |                                                                                |                                                                                                                                                                         |
|---------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Focus</b>  | Automated soil sampling, nutrient testing (NPK), and fast in-field analysis. | Soil penetration measurement, moisture/temperature sensing, and GIS mapping. | Automation of soil collection, real-time analysis, and autonomous navigation to enhance agricultural and environmental assessments. | Robotic exploration with continuous soil moisture sensing and Kriging mapping. | Integrating sensors, wireless communication, and AI to provide accurate soil analysis, especially in hard-to-reach areas.                                               |
| <b>Sensor</b> | NPK sensor, pH sensor, inertial sensor, ultrasonic sensors (for obstacles).  | Penetrometer, force sensor, moisture sensor, temperature sensor, RTK-GPS.    | Uses soil sensors (moisture, temperature, pH), GPS, IMU, environmental sensors, and cameras/LiDAR for navigation.                   | Non-contact soil moisture sensor.                                              | Using a set of sensors capable of measuring soil moisture, soil temperature, air temperature, and air humidity, integrated with a GPS module and wireless communication |

|                             |                                                                                  |                                                                          |                                                                                                            |                                                                      |                                                                                                         |
|-----------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
|                             |                                                                                  |                                                                          |                                                                                                            |                                                                      | technologies.<br><br>mounted on the vehicle to measure soil conditions directly.                        |
| <b>Real-time Monitoring</b> | Instant display of NPK values on LCD and real-time obstacle detection.           | Live data collection, automatic stopping, and real-time GPS positioning. | Provides live tracking, real-time soil data updates, system alerts, and continuous performance monitoring. | Live, on-the-move moisture data collection during field exploration. | Sensor data is sent instantly to the mobile app so users can monitor soil status live.                  |
| <b>Technology</b>           | STM32 controller, auger drill, linear actuator, DC motors, solar-powered system. | BoniRob robot, ROS, linear actuators, Arduino, industrial PC, GIS tools. | UGVs, autonomous navigation, IoT-based sensing, wireless communication, and data processing software.      | Autonomous mobile robot, localization system, Kriging-based mapping. | IoT communication, an ESP32 microcontroller, a GPS module, environmental and soil sensors, and AI-based |

|                           |                                                                                |                                                                           |                                                                                                               |                                                                                |                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
|                           |                                                                                |                                                                           |                                                                                                               |                                                                                | analysis.                                                                                              |
| <b>Data Analytics</b>     | Reads soil NPK levels and converts sensor values into nutrient concentrations. | Analyzing soil compaction, generating maps, and comparing sensor results. | Real-time soil data analysis, multi-sensor fusion, and generation of soil maps; may include machine learning. | Uses Kriging interpolation to estimate moisture distribution across the field. | Collected readings are stored and analyzed to detect soil patterns and support better decision-making. |
| <b>Mobile Application</b> | No                                                                             | Yes                                                                       | Yes                                                                                                           | No                                                                             | Yes                                                                                                    |

This proposed project is different from earlier developed systems since this system makes use of real time soil testing, wireless communications, GPS tracking, and intelligent data analysis all combined together into one smart agriculture system. Different from most studies performed before which made laboratory soil tests or only determined a particular soil property, this new system makes direct measurements of the soil properties in real time in the field using a remote-control excavator equipped with multiple sensors for measuring soil moisture, soil temperature, air temperature, and air humidity, in addition to a GPS module for recording the location of each reading.

Data obtained are associated with geographic locations and shown on a mobile application interface. This makes the soil monitoring process efficient. Moreover, intelligent data analysis would be helpful for agricultural decision making.

### 3 CHAPTER 3: METHODS AND MATERIALS

This chapter provides the overall methodology and technical design for the system of analyzing and monitoring the soil by utilizing the smart vehicle concept. It includes the system architecture, the structure of the mobile vehicle, hardware/software aspects, sensing mechanisms, communication mechanisms, data processing, and other relevant design decisions for the proposed system and the rationale for the selection of various technologies for the architecture and design.

#### 3.1 System Design and Components

This section describes in brief the overall architecture and major constituents of the proposed system. The system architecture is modular, which makes it flexible, expandable, and maintainable. The system consists of hardware elements, which include sensors, actuators, and the mobile unit, and software elements, which carry out functions such as data collection, processing, and control. The hardware and software combined allow for accurate data collection under agricultural field environments.

##### 3.1.1 Block Diagram

The block diagram which describes the full process and components that will be used in the system is shown in figure below.

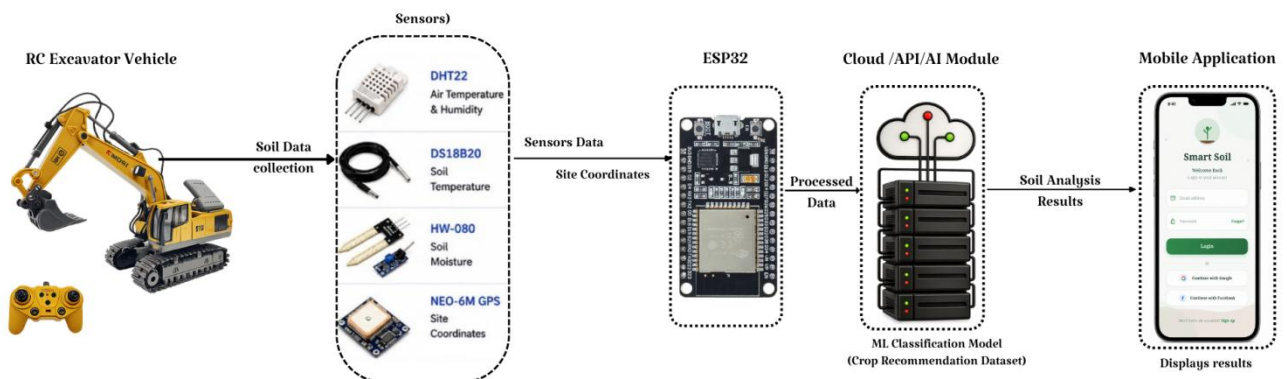


Figure 3.5 :Block Diagram

### 3.1.2 Mobile Application Design

The mobile application is proposed to be the primary user interface component of the proposed intelligent system for soil monitoring. The mobile application is intended to display information on the soil conditions using artificial intelligence. The intelligent system is designed to use the results of the soil test to establish the best crop to cultivate in the region based on the soil characteristics. The mobile application is convenient and offers an easy-to-use interface for users such as farmers and agricultural engineers.

Figure 3.2 shows a detailed design for the application interface, which presents the mechanism for the interaction between the user and the system components.

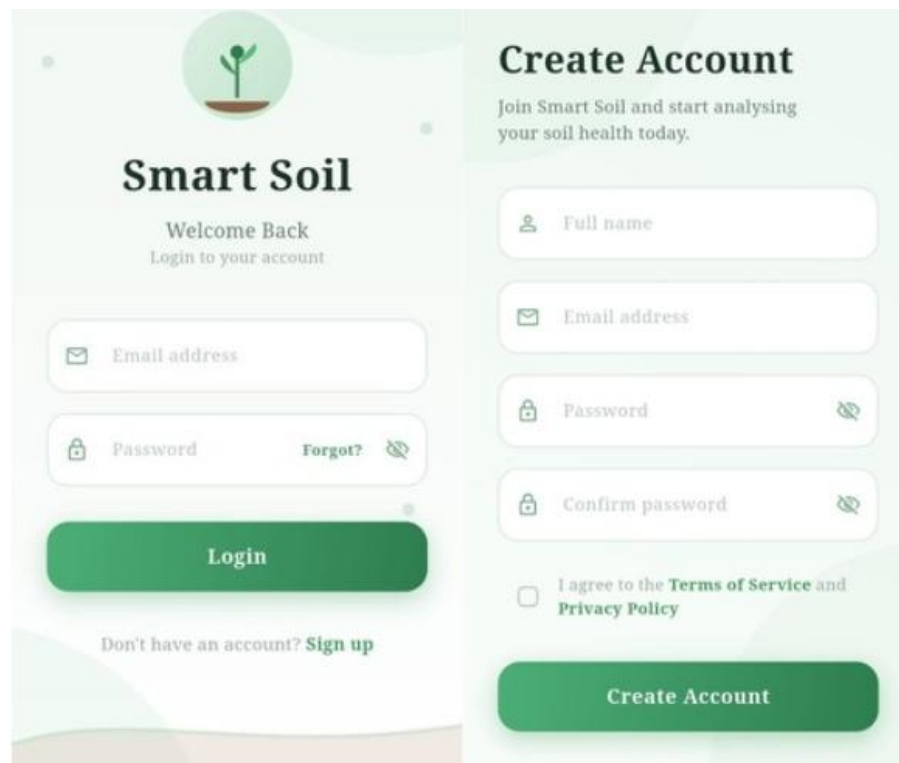


Figure 3.2 :Sample screens of the mobile application design

### 3.1.3 Hardware Components

This section shows the main hardware components of our system.

- **ESP32 Development Board – DEVKIT V1**

ESP32 Development Board DEVKIT V1 ESP32 Development Board DEVKIT V1 is a resourceful microcontroller board employed for IoT technologies. It has a low power consumption rate, hence it is appropriate for real-time monitoring since it contains a dual-core Xtensa LX6 processor. It allows Wi-Fi and Bluetooth connectivity; hence, it is efficient in wirelessly transmitting data as well as for the control of the system. Connection to most sensors can be done with this board, and its portability and numerous GPIO ports make it perfect for environmental and soil monitoring. Hence, ESP32 provides an efficient as well as scalable solution for the proposed smart soil inspection system [9].



*Figure 3.3 :ESP32 [9]*

- **Temperature and Humidity Sensor (DHT22)**

DHT22 sensor is the sensor employed in the proposed system to measure ambient air temperature and relative humidity. This is an accurate digital sensor capable of giving out temperature and humidity data via a digital signal from a single wire interface. The use of this sensor enables the system to have accurate data with wide ranges of measurements as opposed to other inexpensive sensors such as DHT11. In the proposed system, DHT22 sensor was chosen since it is capable of giving out stable environmental data useful in recommending crop type and determining the agricultural environment [10].



*Figure 3.4 :Sensor(DHT22)[10]*

- **DS18B20 Soil Temperature Sensor**

The soil temperature will be measured using DS18B20 digital temperature sensor in the proposed system. This is a digital thermometer which uses 1-Wire communication interface. Through this approach, temperature measurements are made possible reliably by using a single data wire in addition to voltage and ground wires. The device has a programmable resolution and is commonly used in environmental and agricultural applications because of its precision, ease of use, and reliability. In the proposed system, the DS18B20 sensor will be used to measure soil temperature in the field environment as soil temperature is a vital parameter in seed germination, nutrients absorption, and crops growth[11].



*Figure 3.5:Sensor(DS18B20)[11]*

- **Lithium-Ion Battery**

Widely used rechargeable battery which consists of several cells: cathode, anode, current collector, separator, and electrolyte. The electrical current is created when the lithium ions move between cathode and anode and then the electrons move opposite to them in the external circuit. While the battery is charging lithium ions are released by the cathode and received by the anode. The opposite happens when discharging. Figure 3.6 illustrates the operating principle of the lithium-ion battery used in the proposed system [\[12\]](#).



*Figure 3.6 :Lithium-Ion Battery [\[12\]](#)*

- **NEO-6M GPS module**

NEO-6M GPS module is used to identify the geographical position of each soil sample measuring point. This module receives satellite signals and gives geographical coordinates like latitude and longitude.

In the proposed system, the GPS module is interfaced with ESP32 to relate each soil sample reading to its respective geographical position. This will be beneficial for users to monitor soil condition based on their spatial location.

The NEO-6M module was chosen due to its accuracy, affordability, and compatibility with embedded systems and IoT projects[\[13\]](#).



*Figure 3.7 :NEO-6M GPS Module [13]*

## ● RC Excavator Vehicle

A commercially available remote-controlled excavator vehicle is chosen as the mobile vehicle in the proposed system. The excavator vehicle enables mobility across various agricultural sites while transporting sensors and communication modules.

The soil sensor is attached to the excavator arm, which facilitates inserting the soil sensor into the ground while collecting data. The excavator vehicle can be operated via remote control using a wireless remote controller.

The use of an existing excavator vehicle eliminates the need for designing a complex mechanical system, which enables the project to concentrate on soil sensing, wireless communication, and intelligent data processing[14].



*Figure 3.8 :RC Excavator Vehicle [14]*

### ● HW-080 Soil Moisture Sensor

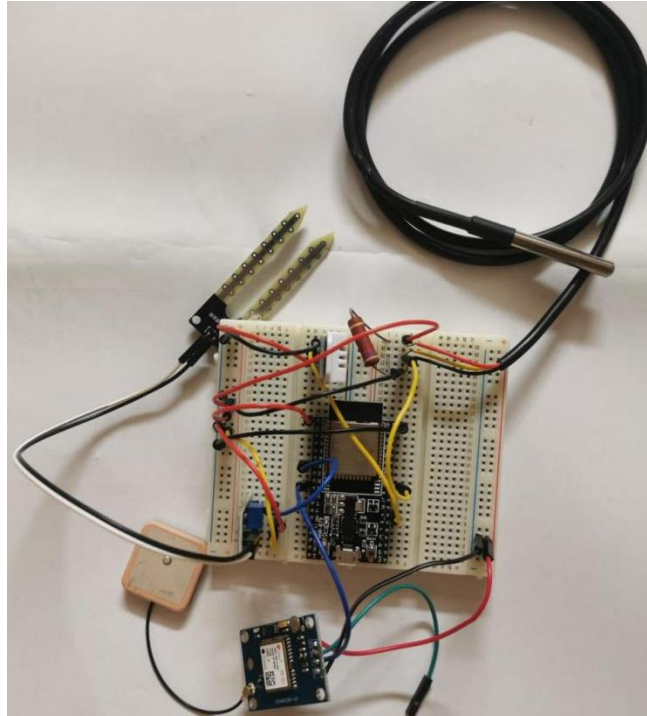
The soil moisture content of the soil was calculated using the HW-080 soil moisture sensor. This is achieved through the detection of the variation in the conductivity between the probes in the sensor after inserting the sensor into the soil based on the moisture content present in the soil. In the suggested system, the HW-080 sensor provides an efficient and cost-effective way of monitoring the soil moisture in real-time. Soil moisture content is one of the main environmental factors that can be monitored in agriculture due to its importance and impact on the availability of water for irrigation and crop planting[15].



*Figure 3.9 :Sensor(HW-080)[15]*

### ✓ Hardware Prototype Setup

The full hardware prototype of the proposed intelligent soil monitoring system was designed using a circuit breadboard setup. The ESP32 development board acted as the main controller and was connected to the soil moisture sensor, DS18B20 temperature sensor, air moisture and temperature sensor, GPS module, as well as other necessary electronic parts needed by the system for proper functioning. This setup allowed the collection of live environmental and soil data and also helped in transmitting the data to the mobile app and Firebase cloud database.



*Figure 3.10 :Hardware prototype setup of the proposed smart soil monitoring system*

### 3.1.4 Dataset Description

The machine learning aspect of the proposed system was designed based on the Crop Recommendation Dataset [\[16\]](#). This dataset was chosen since it holds agricultural and environmental parameters that directly affect the crop production process.

The input features contained in the dataset include the following:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Temperature
- Humidity
- pH
- Rainfall

The output variable involves 22 classes of crops that indicate the best crop to be grown under specific environmental conditions[\[16\]](#).

The dataset was uploaded to Edge Impulse Studio by using the CSV Wizard utility. In the process of importing the dataset, the crop category was designated as the target output label, whereas the other parameters acted as input features for designing a machine learning algorithm.

Prior to training the algorithm, the dataset was analyzed and arranged for adequate feature mapping. The dataset served as a basis for creating an intelligent crop recommendation model that could help make agriculture-related decisions based on soil parameters.

### **3.1.5 AI Model Development**

The AI model was built with an aim of making crop recommendations considering soil and environmental conditions. After creation of the Crop Recommendation Dataset, the process of building the model was carried out in the Edge Impulse Studio.

#### **3.1.5.1 Dataset Selection and Preprocessing**

The Crop Recommendation Dataset was chosen as the foundation for creating the machine learning model. At first, the initial dataset included the following input parameters: Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, and Rainfall, with 22 crop categories as the output variables. The rainfall feature used in the model was based on average annual rainfall data obtained from ArcGIS Hub [\[17\]](#).

The dataset was uploaded to Edge Impulse Studio via the CSV Wizard tool. While importing the dataset, the crop class was designated as the target output, while other parameters were set as input variables for further training.

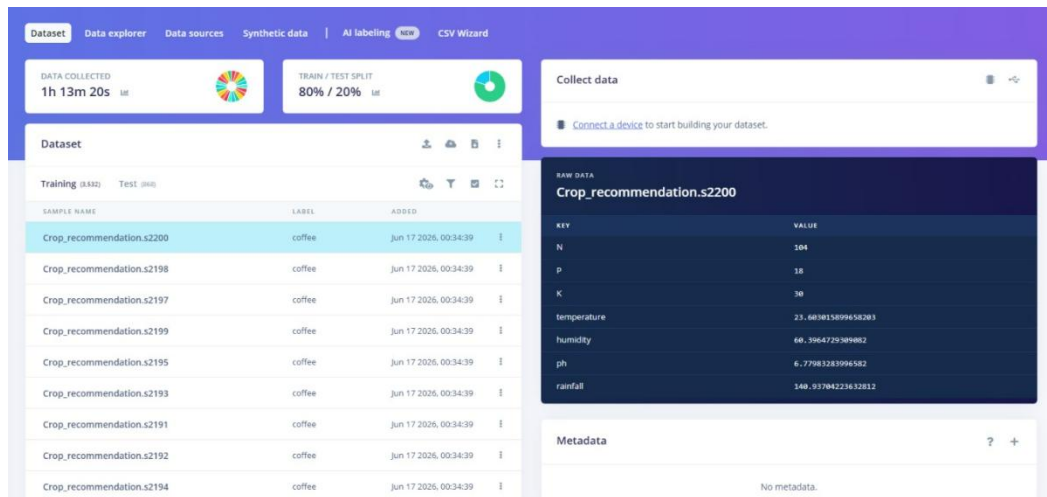


Figure 3.11 :Crop Recommendation Dataset in Edge Impulse Studio

## Feature Selection

At first, only two features, such as temperature and humidity, were used as input features due to the limitation of the hardware. However, the model showed only 65% classification accuracy, which means that only these two features are not enough to create an adequate recommendation system.

Therefore, to increase the accuracy of the model, more agricultural parameters were used as the input features of the model, such as Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, and ph. Rainfall was excluded at first and considered during experimentation[18].

These agricultural parameters were selected because they are among the main indicators used in soil fertility assessment and have a direct influence on crop suitability and productivity under different soil conditions [19].

## Feature Importance Analysis

Feature importance was analyzed to assess the level of influence of each input variable on the crop classification model. It was found out that humidity, rainfall, and potassium (K) were some of the most important variables influencing the crop recommendation decision process. Nitrogen (N), phosphorus (P), temperature, and pH were also some of the factors influencing the classification process.

Assessment of feature importance is essential for identifying the most influential environmental and soil variables in the crop classification process.



*Figure 3.12 :Feature Importance*

## Edge Impulse Configuration

A machine learning model was designed in Edge Impulse Studio with the following specifications:

- Processing Block: Raw
- Learning Block: Classification
- Train Data Distribution: 80%
- Feature Scaling: Standard Scaler (Sklearn)
- Output Classes: 22 crops

Features were scaled in order to ensure that each parameter carried equal weight while training the neural network.

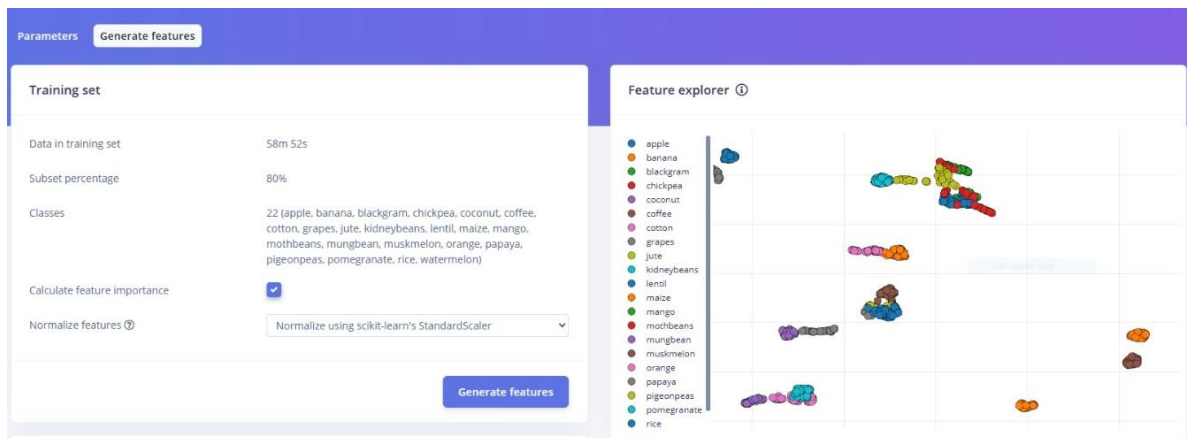


Figure 3.13 :Feature Generation and Visualization

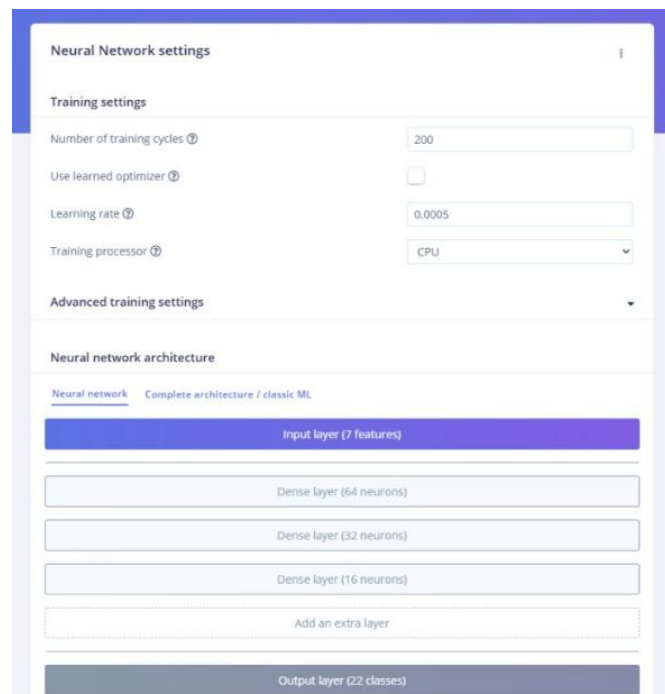
## Neural Network Structure

The MLP (Multi-Layer Perceptron) neural network model, which is an example of ANN (Artificial Neural Network), was employed for classification of crops.

Structure of the artificial neural network is illustrated below:

- Input Layer: 7 Input Features
- Hidden Layer 1: 64 Neurons
- Hidden Layer 2: 32 Neurons
- Hidden Layer 3: 16 Neurons
- Output Layer: 22 Crop Classes

The model was optimized by using the Adam optimizer with the learning rate set to 0.0005.



The screenshot displays a web-based interface for configuring a neural network. It is divided into two main sections: 'Neural Network settings' and 'Neural network architecture'.

**Neural Network settings:**

- Training settings:**
  - Number of training cycles: 200
  - Use learned optimizer: ☐
  - Learning rate: 0.0005
  - Training processor: CPU
- Advanced training settings:** (Collapsed)

**Neural network architecture:**

- Neural network (selected) / Complete architecture / classic ML
- Input layer (7 features)
- Dense layer (64 neurons)
- Dense layer (32 neurons)
- Dense layer (16 neurons)
- Add an extra layer
- Output layer (22 classes)

Figure 3.14 :Neural Network Architecture and Training Settings

## **Model Inference and Output Generation**

The developed crop recommendation model has been designed based on a supervised machine learning classification algorithm, which has been trained on the labeled data available from the Crop Recommendation Dataset. In the process of making a prediction or recommendation, the model takes into consideration seven input variables, such as nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH value, and rainfall. These inputs result in the production of probabilities of all 22 crops classes included in the dataset. The crop class having the highest probability is chosen as the output or the final recommendation of the model.

## **Model Optimization**

Various training setups were tested to optimize the classification accuracy of the neural network. The initial models provided lower validation accuracies compared to the required level. Adding an extra hidden layer in the neural network structure was found to increase its ability to separate crops.

Normalization of the input features and extended training cycles were utilized to improve convergence and classification accuracy.

## **Model Deployment on ESP32**

Once the crop recommendation algorithm had been trained and validated in the Edge Impulse Studio environment, TinyML methodology was used to deploy the machine learning model on the ESP32 microcontroller chip. In this way, it was possible to ensure that the classification algorithm could be executed on the embedded system directly, without any need for online processing of the data on the cloud side. Deploying the machine learning model on the ESP32 chip enables the system to perform local processing of the input parameters related to agriculture and produce the crop recommendation accordingly.

## Model Performance

The resulting neural network showed an accuracy of 97.9% with a loss of 0.05 during the validation process. The majority of crop types were able to provide more than 0.90 as F1-score which shows good abilities of classification. Lower accuracy of a number of crops such as Rice, Jute, Lentil and Pigeon Peas has been observed due to similar requirements concerning environment and soil type.



*Figure 3.15 :Crop Recommendation Model Performance*

Additional metrics were used for measuring the performance of the trained model. The results show that the precision was 0.98, recall was 0.98, and the F1 score was 0.98. Data Explorer graph shows that most crop classes were well-separated from each other, which suggests good classification and

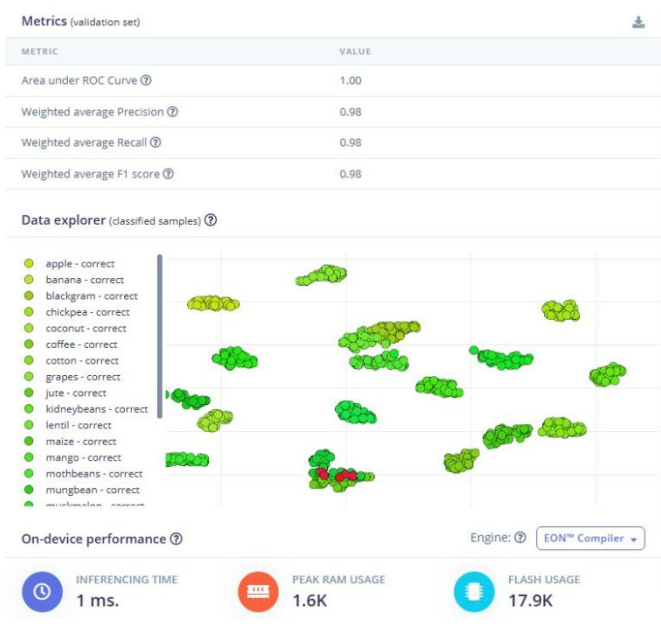


Figure 3.16: Validation Metrics and Data Explorer

**Confusion matrix** (validation set)

|             | F1-SCORE | PRECISION | RECALL |
|-------------|----------|-----------|--------|
| APPLE       | 1.00     | 1.00      | 1.00   |
| BANANA      | 1.00     | 1.00      | 1.00   |
| BLACKGRAM   | 0.94     | 1.00      | 0.89   |
| CHICKPEA    | 1.00     | 1.00      | 1.00   |
| COCONUT     | 0.97     | 1.00      | 0.94   |
| COFFEE      | 0.96     | 0.92      | 1.00   |
| COTTON      | 1.00     | 1.00      | 1.00   |
| GRAPES      | 1.00     | 1.00      | 1.00   |
| JUTE        | 0.88     | 1.00      | 0.73   |
| KIDNEYBEANS | 1.00     | 1.00      | 1.00   |
| LENTIL      | 0.96     | 0.92      | 1.00   |
| MAIZE       | 1.00     | 1.00      | 1.00   |
| MANGO       | 1.00     | 1.00      | 1.00   |
| MOTHBEANS   | 1.00     | 1.00      | 1.00   |
| MUNGBEAN    | 1.00     | 1.00      | 1.00   |
| MUSKMELON   | 1.00     | 1.00      | 1.00   |
| ORANGE      | 1.00     | 1.00      | 1.00   |
| PAPAYA      | 1.00     | 1.00      | 1.00   |
| PIGEONPEAS  | 1.00     | 1.00      | 1.00   |
| POMEGRANATE | 0.96     | 0.93      | 1.00   |
| RICE        | 0.84     | 0.79      | 1.00   |
| WATERMELON  | 1.00     | 1.00      | 1.00   |

Figure 3.17 :Confusion Matrix

## Strategy for the Model Deployment

In order to deploy the model in practice, the use of the sensors which can collect the environmental data such as soil temperature, air humidity, soil moisture level, and GPS location is being made. The additional parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), pH, and rainfalls can be calculated based on the agricultural and climatological data.

### 3.2 Design Specifications ,Standards and constraints

The soil analysis and monitoring system offered in the project can be applied in smart agriculture through the use of remotely operated excavator equipment equipped with sensors, communication, and analysis devices. In other words, it can be used to collect data about different parameters of soil, including moisture, temperature, pH, EC, and NPK values.

The collected information will be analyzed by means of the ESP32 microcontroller and transmitted wirelessly using Wi-Fi and Bluetooth technologies to the mobile application. In addition, the system will be able to detect geographical coordinates of each soil sample in order to ensure its proper analysis.

The proposed system meets IoT requirements, such as wireless connection, real-time monitoring, energy efficiency, and combining hardware and software elements. The system may be upgraded with new sensors and additional analysis of the data.

However, there are certain limitations that should be considered while developing the system. On the one hand, the performance of the system is dependent on the computing power, reliability of the connection, and energy consumption. On the other hand, there are certain environmental limitations related to soil type, climate conditions, and other factors.

### 3.3 Specifications

The suggested soil analysis and monitoring system consists of functional and non-functional specifications aimed at ensuring that it operates efficiently and reliably in the agricultural environment. The system is capable of measuring some important soil characteristics such as soil moisture, temperature, and measuring air moisture and temperature.

The obtained data is analyzed by the ESP32 micro-controller and transferred to a mobile application through the wireless connection via Wi-Fi and Bluetooth technologies. Moreover, the system contains a GPS module to correlate soil measurements with their geographical coordinates.

It should be noted that the suggested system is practical and scalable and can be used for smart agriculture purposes.

#### **Functional Requirements**

##### **1. Real-Time Soil Data Collection**

- The system shall periodically measure soil parameters such as soil moisture.
- The system shall be able to transmit soil and environmental data to the mobile application through Wi-Fi.

##### **2. Location-Based Measurement**

- The system shall be able to calculate the precise geographical position of every soil sample through the GPS module.
- Every soil reading should be associated with the respective location and shown on the mobile app.

##### **3. Mobile Vehicle Operation**

- The system will enable the robotic vehicle to move to other locations to inspect the soil.

- The robotic arm is deployed to place the sensing unit close to or into the soil for data collection, depending on the sensor type and measurement needs, then returns to its original position.

### **Non-Functional Requirements**

#### **1. Security**

- The system should ensure the secure transmission of soil information and location details in order to prevent unauthorized access.

#### **2. Ease of Use**

- The mobile app should be user-friendly and should enable the user to quickly display the soil analysis report and the location details.

#### **3. System Reliability**

- The system will function reliably in outdoor conditions and will carry out soil analyses without interruption.

#### **4. Scalability**

- The system should support the addition of more sensors or advanced AI soil analysis in the future.

#### **5. Performance**

- The system shall provide timely and accurate soil analysis results, with real-time updates displayed on the mobile application.

### **ESP32 Resource Utilization**

Once the embedded application code was compiled with the help of the Arduino IDE, memory usage by the ESP32 module was analyzed in order to determine whether the system can be effectively executed in the provided hardware environment. The compiled code took

up 366,906 bytes of flash memory, which equals 27% of the total program storage space (1,310,720 bytes). Moreover, global variables took up 23,444 bytes of RAM, which is 7% of the total RAM memory (327,680 bytes), leaving 304,236 bytes of RAM for local variables and other operations. Thus, it becomes evident that the developed embedded application works safely within memory limits and there is enough memory to implement future additions to the system.

### **3.4 Constraints**

#### **1. Hardware Constraints**

- The ESP32 Development Board has limited processing power and memory compared to advanced industrial controllers.
- The number of sensors and actuators is limited by the available GPIO pins on the ESP32.
- The robotic arm and sensors are constrained by the mechanical strength and weight capacity of the vehicle chassis.

#### **2. Software Constraints**

- Cloud services and databases may have limitations on data storage and access frequency depending on the service plan used.
- AI-based soil analysis accuracy depends on the availability and quality of collected data.

#### **3. Network Constraints**

- Real-time data transmission and remote monitoring depend on the availability and stability of the Wi-Fi connection.
- Network delays or interruptions may affect real-time updates in the mobile application.

#### **4. Environmental Constraints**

- The performance of the system could be affected by the soil type, moisture, as well as temperature and humidity in the external environment.
- Issues such as dust, vibration, and uneven farming terrain could affect the accuracy, precision, as well as movement of the excavator that is controlled remotely.
- Communication of sensors with the remote station could also be affected by environmental issues.

### 3.5 Design Alternatives

Several design approaches have been considered in the development process of this project. One alternative could be an Arduino board with external Wi-Fi and Bluetooth modules. However, this solution was not taken because the ESP32 board provides Wi-Fi and Bluetooth capabilities internally, thus minimizing the inclusion of several other supporting components in the system design, reducing the complexity of hardware for power consumption, and improving communication reliability. The usage of ESP32 minimizes wiring and integration problems as compared to using multiple external communication modules with Arduino.

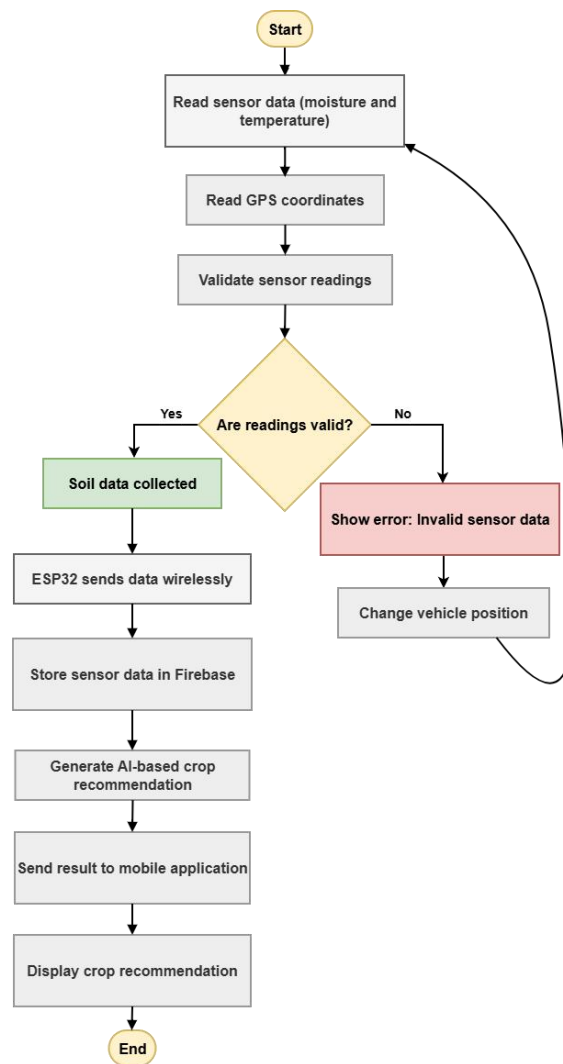
Another possible alternative would have been to use a number of different sensors to measure directly other soil parameters such as pH and nutrient related values. This option was however not adopted because some of these sensors were difficult to find locally during the project development period. Thus, the available sensors for soil moisture, soil temperature, air temperature and air humidity were used to develop the proposed system and machine learning was used to help estimate other soil-related parameters and crop recommendation.

The proposed system employs a sensing set up with DHT22, DS18B20 and HW-080 sensors, a NEO-6M GPS module and an ESP32 microcontroller in place of the ones mentioned above. This design was selected as it could provide the required real-time measurements for the project and simultaneously reduce the hardware complexity, ease of integration and cost. Moreover, the system is extended with the help of machine learning for the estimation of other soil-related parameters and crop recommendation.

## 3.6 System Analysis and Optimization

### 3.6.1 Flowchart

The flow chart is as shown in the figure below. It represents the workflow of our full project. workflow of our full project.



*Figure 3.18 :Flowchart*

### 3.6.2 Use Case daigram

To illustrate the primary interactions among users and system, we have drawn a use case diagram, as depicted in Figure 3.19 .

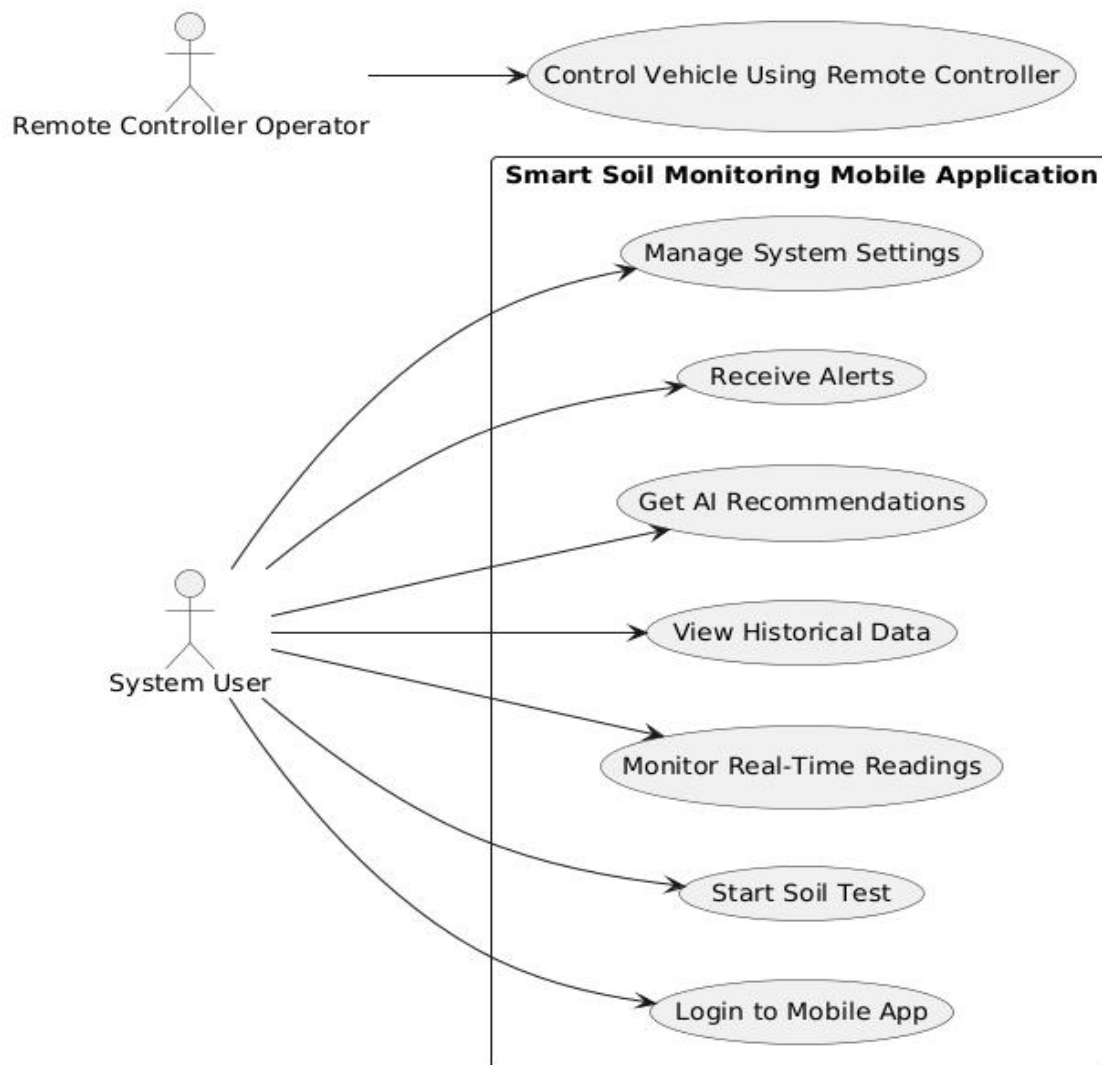


Figure 3.19 :Use Case Daigram

This table depicts major use cases of the intelligent analysis and monitoring system. It shows the interaction of users (user and administrator) with the functions of the system. The use cases include real-time monitoring of the soil data, alert messages on abnormal readings, user login, remote control of vehicles, start test of soil, generation of intelligent recommendations based on AI, and display of historical data. Further, prerequisites for each use case and the required steps to perform or implement it have also been provided in the said table, which makes the understanding of system behavior or functions clear and organized.

*Table 2 :Use case*

| Use Case                      | Actors           | Description                                                                                                     | Preconditions                                                                                                      | Steps                                                                                                                                                                                                                        |
|-------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Monitor Real-Time Data</b> | User,<br>Manager | Users view real-time soil measurements such as moisture and temperature ,ph,NPK through the mobile application. | <ul style="list-style-type: none"> <li>• Sensors are active.</li> <li>• System is connected to the app.</li> </ul> | <ol style="list-style-type: none"> <li>1. Sensors collect soil data.</li> <li>2. The system receives the data.</li> <li>3. The system processes the readings.</li> <li>4. Real-time data is displayed in the app.</li> </ol> |

|                                 |                  |                                                                                                 |                                                                                                                                |                                                                                                                                                                                                                          |
|---------------------------------|------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Receive Alerts</b>           | User,<br>Manager | Notifies users when abnormal soil readings or system issues occur.                              | <ul style="list-style-type: none"> <li>• System is monitoring sensor data.</li> <li>• Alert thresholds are defined.</li> </ul> | <ol style="list-style-type: none"> <li>1. System continuously checks sensor values.</li> <li>2. If abnormal values are detected, an alert is generated.</li> <li>3. Notification is sent to the user/manager.</li> </ol> |
| <b>User Login</b>               | User,<br>Manager | Users log in to access system features and results.                                             | User has a registered account.                                                                                                 | <ol style="list-style-type: none"> <li>1. Users enter login credentials.</li> <li>2. System verifies the credentials.</li> <li>3. Access is granted based on the user role.</li> </ol>                                   |
| <b>Control Vehicle (Remote)</b> | User             | Users manually control the vehicle using a remote controller to reach the target soil location. | <ul style="list-style-type: none"> <li>• Vehicle power is ON.</li> <li>• Remote controller is connected.</li> </ul>            | <ol style="list-style-type: none"> <li>1. User turns on the vehicle and remote.</li> <li>2. Users drive the vehicle to the target area.</li> <li>3. User stops the vehicle at the sampling point.</li> </ol>             |

|                             |               |                                                                      |                                                                                                                                                                |                                                                                                                                                                                                                         |
|-----------------------------|---------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Start Soil Test</b>      | User          | Starts the soil testing process to collect sensor measurements.      | <ul style="list-style-type: none"> <li>• Vehicle is positioned correctly.</li> <li>• Sensors are operation.</li> </ul>                                         | <ol style="list-style-type: none"> <li>1. User starts the soil test.</li> <li>2. Sensors take soil measurements.</li> <li>3. Data is sent to the system.</li> <li>4. Results are prepared for display.</li> </ol>       |
| <b>AI Recommendations</b>   | User, Manager | AI analyzes soil measurements and provides recommendations and tips. | <ul style="list-style-type: none"> <li>• Soil measurements are available.</li> <li>• AI model is active.</li> <li>• Air measurements are available.</li> </ul> | <ol style="list-style-type: none"> <li>1. System sends measurements to the AI model.</li> <li>2. AI analyzes the data.</li> <li>3. Recommendations are generated.</li> <li>4. Tips are displayed in the app.</li> </ol> |
| <b>View Historical Data</b> | User, Manager | Users view and analyze previous soil test results.                   | Historical data is stored in the system.                                                                                                                       | <ol style="list-style-type: none"> <li>1. User selects a date or record.</li> <li>2. System retrieves historical data.</li> <li>3. Data and recommendations are displayed.</li> </ol>                                   |

### 3.6.3 Activity Diagram

Figure 3.20 shows how the system works in detail when the field operator or manager interacts with the Smart Soil Analysis mobile app. First, after the application has started, the field operator or the manager logs into the application. If the validation passes, it takes the user to the main interface of the application.

The process starts with the soil test for the field operator, in which the system triggers the sensors on and collects data from the soil in real time. Through this, the real-time readings monitoring activity displays the data. During the test, the AI engine is invoked implicitly to analyze the data collected and make some intelligent recommendations. With these recommendations, the field operator can remotely control the vehicle for several adjustments in its movement or excavator arm position. In case there are any abnormal readings detected, immediate alerts are shown to the user by the system. All the readings taken and results analyzed are stored in the database; therefore, the field operator can optionally view historical data or continue doing further tests.

For the manager, on the other hand, workflows involve system monitoring and management. Immediately after his logon to the website, an administrator can view real-time readings of parameters, review historical data, operate and configure system settings, and approve user accounts. Just like in a field operator, the alert is automatically generated when critical conditions are met, and all updates and readings are logged in the database.

In either of the two cases, either the user can continue operating or terminate the session, thus completing one full operating cycle.

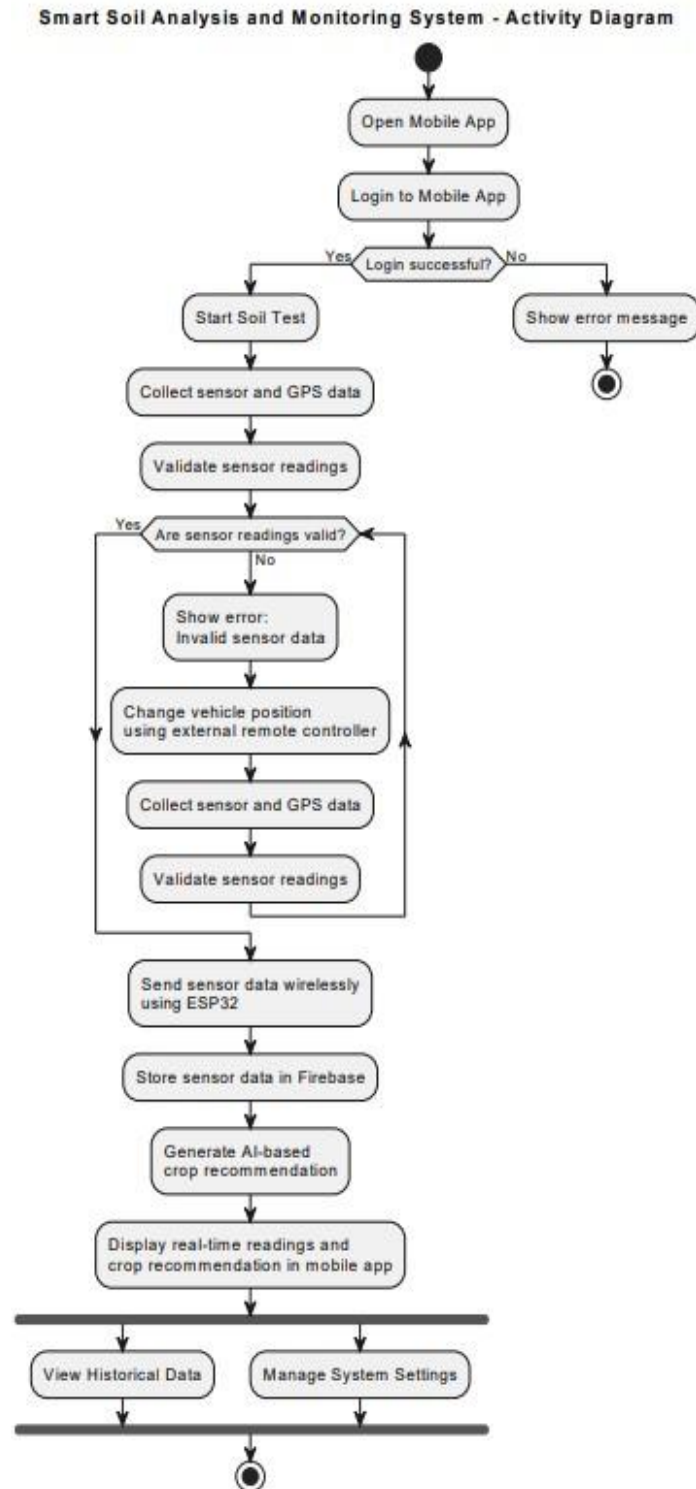


Figure 3.20 :Activity

### 3.6.4 Sequence Diagrams

This diagram in Figure 3.21 shows the overall workflow of the entire system, including the manner to control the vehicle, collect, process, and store the data regarding the soil, together with the manner to display the results under different circumstances.

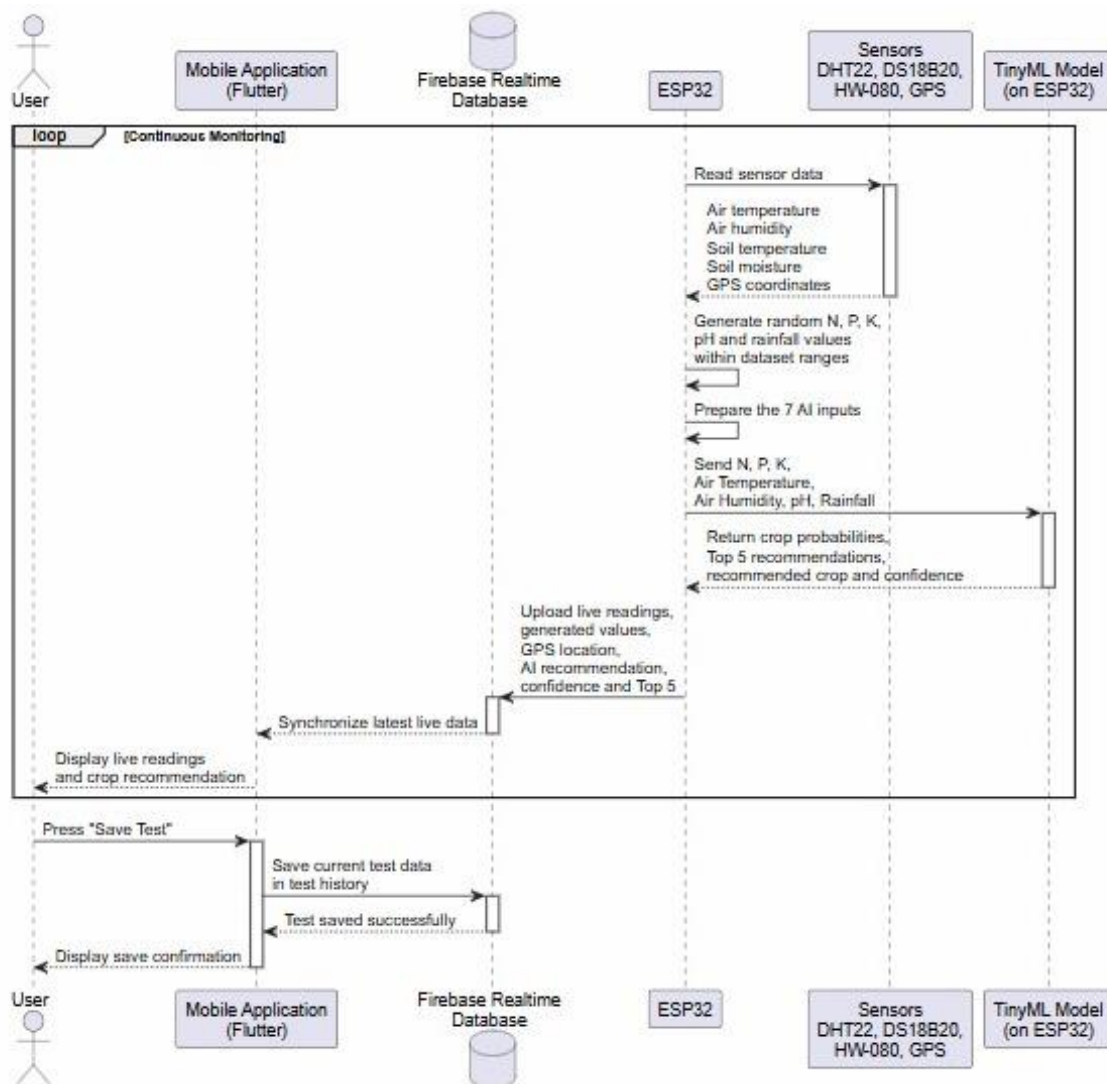


Figure 3.21 :Full Process Sequence Diagram

The diagram above in Figure 3.22 indicates how user authentication is carried out and what is required in order for the user to use the feature provided in the system.

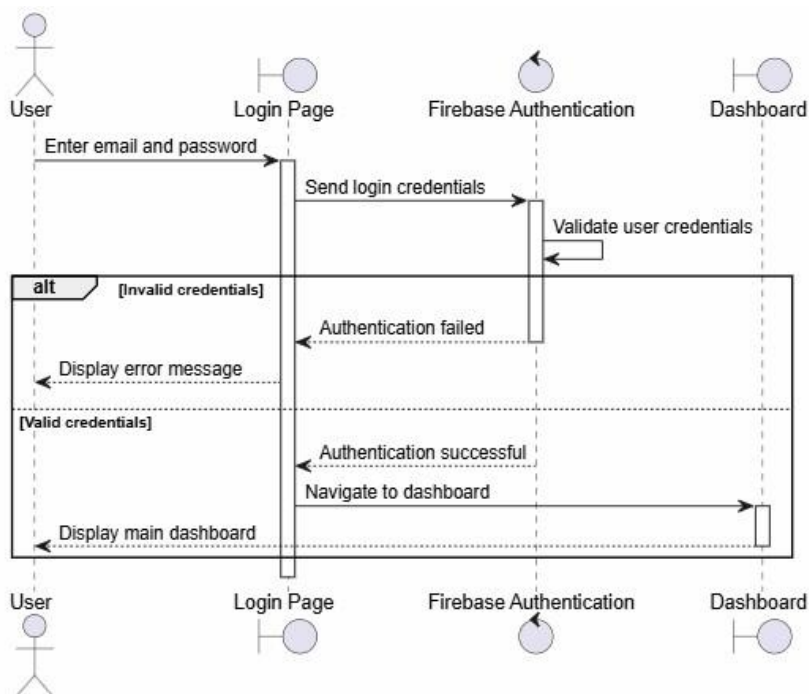


Figure 3.22 :Login Sequence Diagram

Figure 3.23 is a diagram indicating a sequence of operations, starting from the soil testing process, accessing sensor information, an artificial intelligence model, and the display of information.

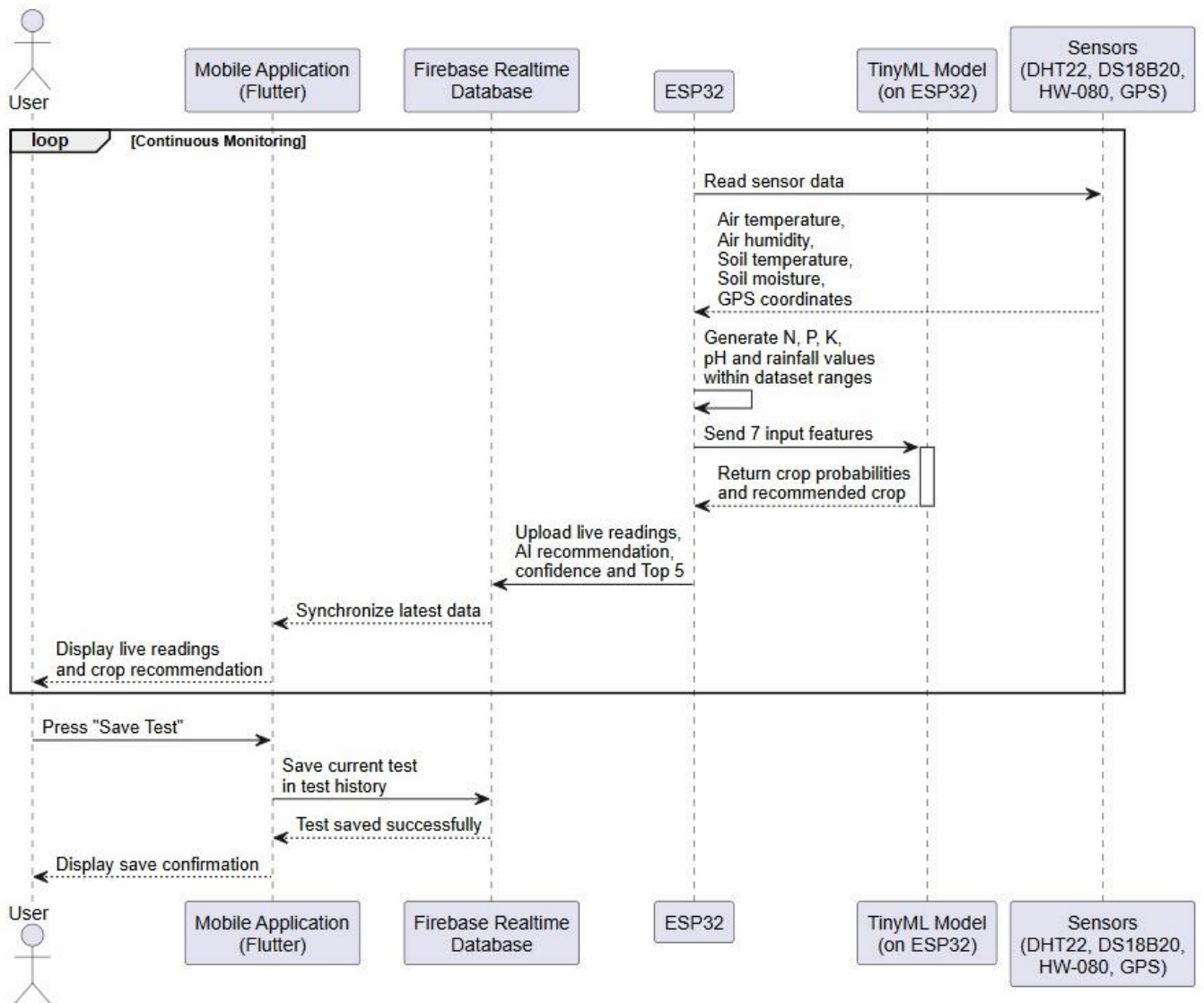


Figure 3.23 :Start Soil Test Sequence Diagram

## 4 CHAPTER 4: RESULTS AND DISCUSSIONS

### 4.1 Results

#### ● Sensor Readings and Data Acquisition

The developed smart soil monitoring system was subjected to the test process to confirm its capability of acquiring environmental and soil-related data in real time. The designed prototype successfully acquired the data provided by the connected sensing units and presented it on the serial monitor screen before passing the data further to the software platform.

The following parameters were acquired during the testing process: air temperature and air humidity via the use of DHT22 sensor; soil temperature via the use of DS18B20 sensor; and soil moisture via the use of HW-080 soil moisture sensor. The aforementioned readings constituted the core of the entire monitoring process and served as a basis for evaluating the functionality of the sensing subsystem.

The serial output showed that ESP32 was capable of reading the connected sensors properly and in real time. The acquired data was constantly updated and ready for transferring to the mobile app and database system.

```
16:59:39.107 -> =====
16:59:39.107 -> Air Temperature : 26.5 C
16:59:39.107 -> Air Humidity : 65.8 %
16:59:39.107 -> Soil Temperature: 25.81 C
16:59:39.107 -> Soil Moisture : 3 %
16:59:39.107 -> Moisture Raw : 4016
16:59:39.107 ->
16:59:39.107 -> GPS : No Fix Yet
```

*Figure 4.1 :Real-Time Sensor Readings on the Serial Monitor*

## ● GPS Data and Location Tracking

The GPS module was incorporated in the system for providing location information along with the measurements that have been taken. While operating the system, the ESP32 was receiving the geographical coordinates from the GPS module and connecting it with the data provided by the sensors. Thus, it was possible for the suggested system to work on the principle of location-based monitoring.

Incorporation of the GPS information makes the system more practical, as it can be applied in agriculture where conditions of the environment can differ depending on the location. It is possible for the system to conduct monitoring in an organized manner.

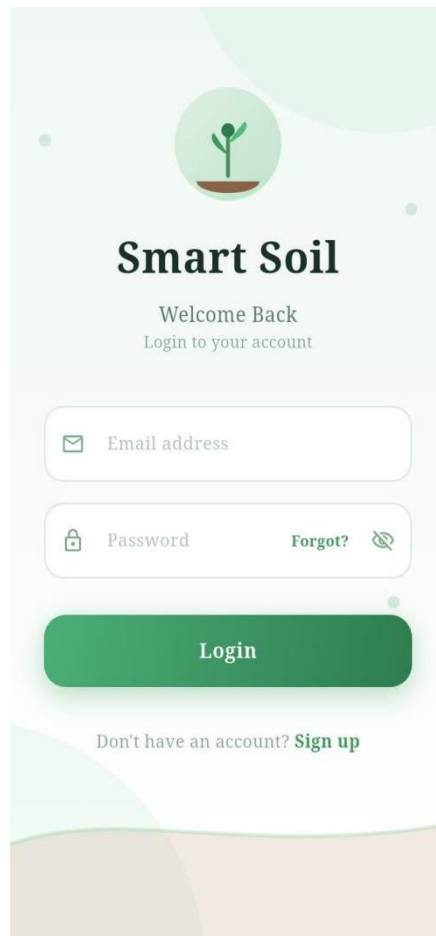
## ● Mobile Application Results

Mobile application served as an interface of the proposed system that was tested during the process. It successfully showed the results of environment and soil parameters obtained by the hardware module and thus provided an opportunity to observe the field status easily.

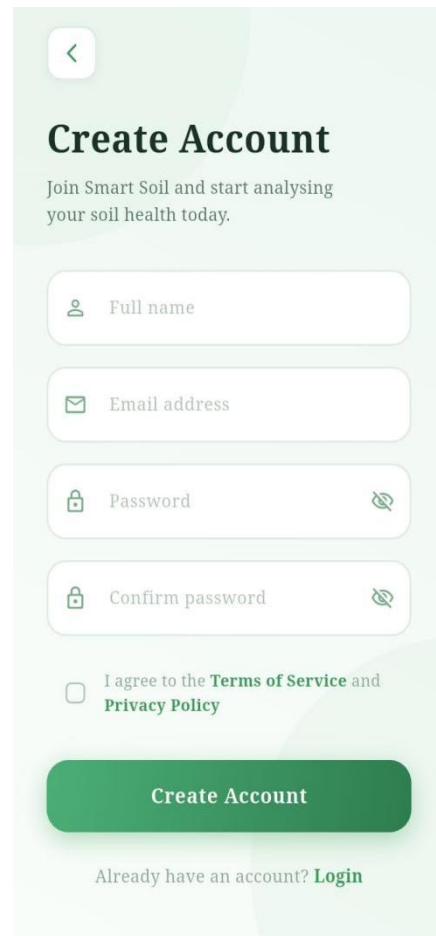
The app has been developed in order to show sensor data clearly and provide an easy way to interact with the system. Moreover, the mobile application made it easier for users to see current data without any physical interaction with the embedded hardware device. Successful

visualization of the sensor readings in the mobile application proved correct connection between ESP32, the database, and software interface.

- **Figures of the Mobile Application Interfaces**



*Figure 4.2 :Login screen*



*Figure 4.3 :Registration screen*

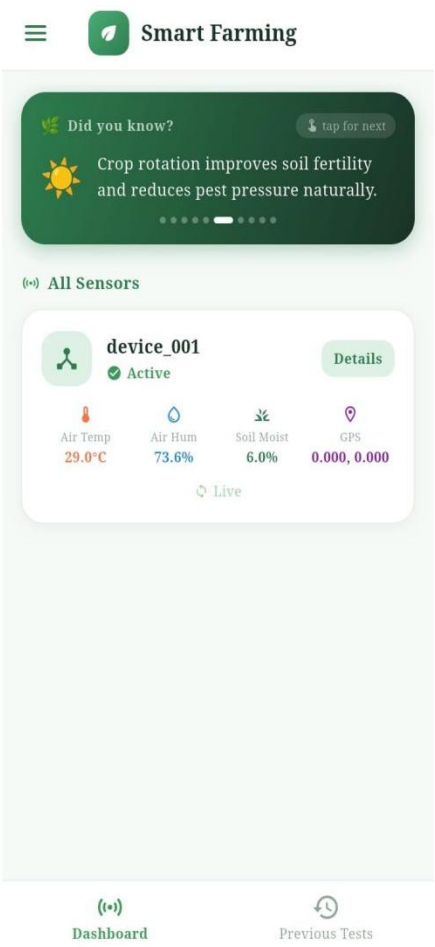


Figure 4.4 :Dashboard screen

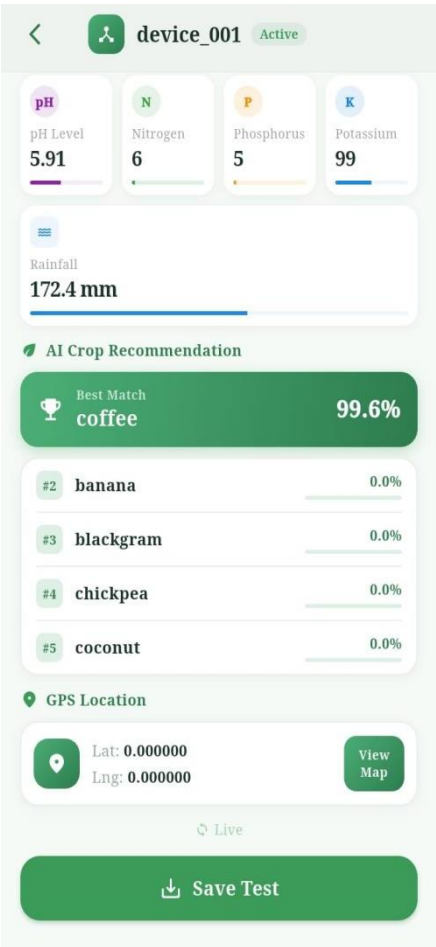


Figure 4.5 :AI soil analysis screen

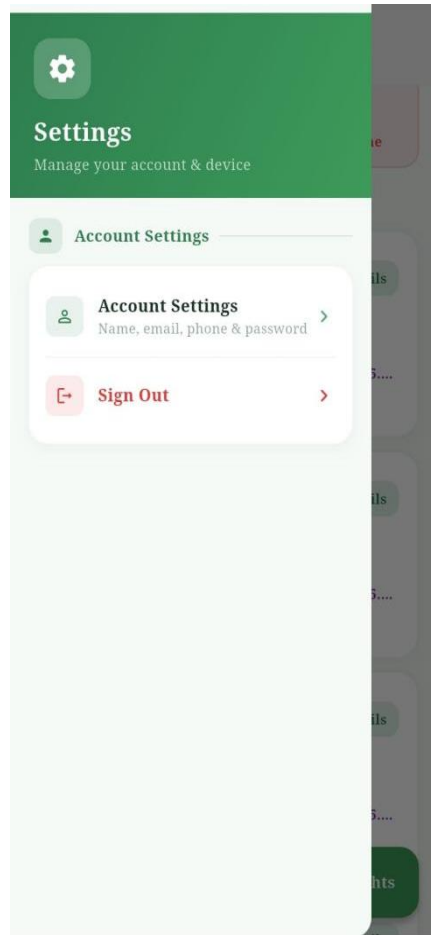


Figure 4.6 :Account settings screen

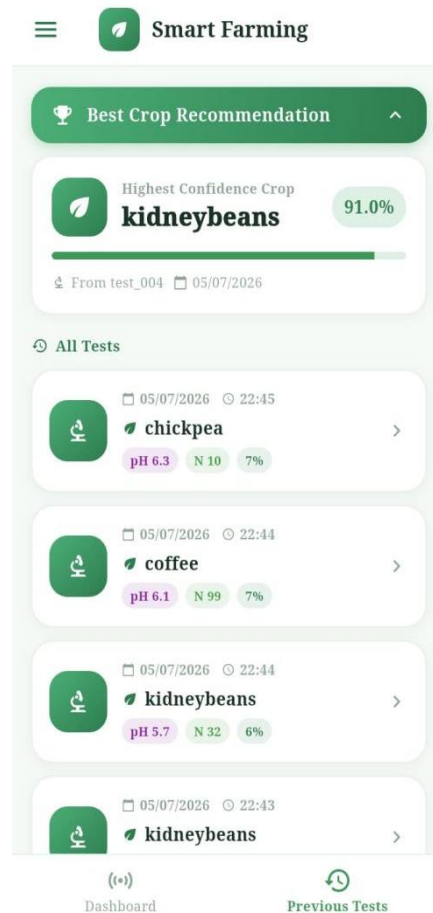


Figure 4.7 :Crop Recommendation Result

## ● Firebase Database Results

Firestore was adopted as the cloud database that stored the readings of the sensors that were taken by the system. During the testing process, it was possible to transfer the readings in real time from the ESP32 and save them in Firestore. This allowed the app to get the readings of the environment and soil.

Firestore provided more flexibility to the system as data could be saved, accessed, and synchronized between the hardware platform and the mobile application.

```
https://smartsoil-e479d-default-rtdb.firebaseio.com
└── https://smartsoil-e479d-default-rtdb.firebaseio.com
    └── devices
        └── device_001
            └── live
                ├── airHumidity: 73.6
                ├── airTemperature: 29
                ├── confidence: 99.60938
                ├── cropRecommendation: "coffee"
                ├── latitude: 0
                ├── longitude: 0
                ├── nitrogen: 6.181212
                ├── ph: 5.911325
                ├── phosphorus: 5
                ├── potassium: 99.47163
                ├── rainfall: 172.4467
                ├── soilMoisture: 6
                ├── soilTemperature: 27.25
                ├── status: "active"
                ├── top5
                ├── updatedAt: 175
                └── tests
```

Figure 4.8 :Sensor readings stored in Firebase database

### ● **AI Recommendation Integration with Firebase**

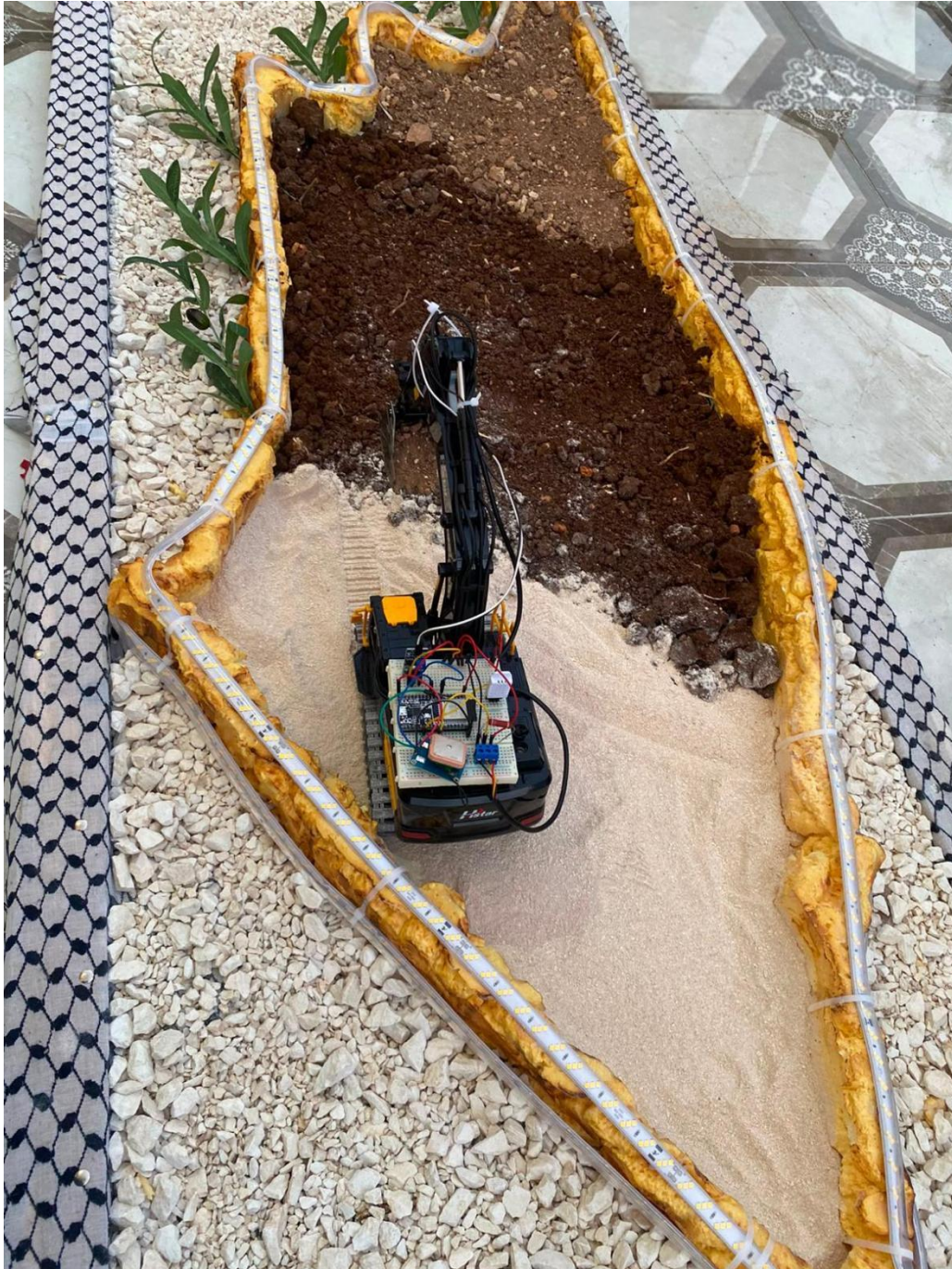
When the model gives an output for the crop recommendation, the crop recommendation prediction along with the sensor reading from the device is sent to the Firebase Realtime Database. The sensor readings consist of air temperature, air humidity, soil temperature, soil moisture, and the location GPS coordinates. Thus, after obtaining the crop recommendation prediction along with the sensor reading, the combined dataset is sent to Firebase Realtime Database, allowing remote monitoring using the mobile application interface.

### ● **System Integration and Overall Operation**

Integration of the entire system was achieved using interactions between all the hardware and software elements. The sensors detected environmental parameters and the soil conditions, the ESP32 gathered the data and sent it, the GPS detected the location of the system, Firebase stored the data received from the hardware, and the mobile app presented the results to the user.

It has been proven that the suggested system is able to implement the whole process of soil monitoring from gathering data to presenting it on the mobile app. Therefore, the proposed system can be used as a smart soil monitoring system.

- **Final Prototype of the Smart Soil Analysis and Monitoring System**



*Figure 4.9: Final Smart Vehicle Prototype*

## 4.2 Discussion

The acquired results prove that the developed solution is able to perform real-time monitoring of soil and environment utilizing low cost sensors and embedded communication technologies. The combination of sensors, wireless communications, cloud database and mobile application proves the feasibility of the developed solution for the purposes of agriculture monitoring applications.

One of the key advantages of the solution is the simplicity and low cost in comparison with the existing solutions for monitoring. Besides, the usage of the mobile application and cloud database adds user-friendly features to the solution. Nevertheless, there are some limitations of the solution such as sensitivity to the sensor calibration, environmental conditions and network connection.

Thus, the results obtained prove that the solution is based on the solid foundation and is able to be used in agriculture monitoring tasks. Moreover, it can be improved through additional sensors utilization, data analysis and large-scale field testing.

## 5 CHAPTER 5: PROJECT MANAGEMENT

### 5.1 Tasks, Schedule and Milestones

In this section we have created a project management schedule and Gantt chart using Microsoft Project. We identified our main and sub-tasks and their sequence, as well as estimated resources and task durations. We have identified important milestones to help track progress and ensure we move toward our final goal.

|                                            |                |                     |                     |          |                         |
|--------------------------------------------|----------------|---------------------|---------------------|----------|-------------------------|
| <b>1 Initiating Phase</b>                  | <b>23 days</b> | <b>Mon 20/10/25</b> | <b>Wed 19/11/25</b> |          |                         |
| 1.1 Identify Team Members                  | 1 day          | Mon 20/10/25        | Mon 20/10/25        |          | Tamara,Aya,Suha         |
| 1.2 assign Project Supervisor              | 3 days         | Tue 21/10/25        | Thu 23/10/25        | 2        | M.Obaid                 |
| <b>1.3 Choose Project Idea</b>             | <b>12 days</b> | <b>Fri 24/10/25</b> | <b>Mon 10/11/25</b> |          |                         |
| 1.3.1 Identify Real-World Problems         | 7 days         | Fri 24/10/25        | Mon 03/11/25        | 3        | Tamara,Aya,Suha         |
| 1.3.2 Meeting with Supervisor for Decision | 3 days         | Tue 04/11/25        | Thu 06/11/25        | 5        | Tamara,Aya,Suha,M.Obaid |
| 1.3.3 Prepare Project Proposal             | 2 days         | Fri 07/11/25        | Mon 10/11/25        | 6        | Tamara,Aya,Suha         |
| 1.3.4 Proposal Approval                    | 0 days         | Mon 10/11/25        | Mon 10/11/25        | 7        | M.Obaid                 |
| <b>1.4 Project Introduction</b>            | <b>7 days</b>  | <b>Tue 11/11/25</b> | <b>Wed 19/11/25</b> |          |                         |
| 1.4.1 Problem Statement and Purpose        | 7 days         | Tue 11/11/25        | Wed 19/11/25        | 8        | Aya                     |
| 1.4.2 Project Design Objectives            | 7 days         | Tue 11/11/25        | Wed 19/11/25        | 8        | Tamara                  |
| 1.4.3 Intended Outcomes and Deliverables   | 7 days         | Tue 11/11/25        | Wed 19/11/25        | 8        | Suha                    |
| 1.4.4 Chapter 1 Approval                   | 0 days         | Wed 19/11/25        | Wed 19/11/25        | 10,11,12 | M.Obaid                 |

Figure 5.1 :A.Project Management Schedule

|                                         |                |                     |                     |          |                 |
|-----------------------------------------|----------------|---------------------|---------------------|----------|-----------------|
| <b>2 Planning Phase</b>                 | <b>59 days</b> | <b>Mon 20/10/25</b> | <b>Thu 08/01/26</b> |          |                 |
| <b>2.1 Related Work</b>                 | <b>23 days</b> | <b>Thu 20/11/25</b> | <b>Mon 22/12/25</b> |          |                 |
| 2.1.1 Search for Similar Applications   | 16 days        | Thu 20/11/25        | Thu 11/12/25        | 13       | Tamara,Aya,Suha |
| 2.1.2 Review Relevant Literature        | 16 days        | Thu 20/11/25        | Thu 11/12/25        | 13       | Tamara,Aya,Suha |
| 2.1.3 Generate Comprehensive Comparison | 7 days         | Fri 12/12/25        | Mon 22/12/25        | 16,17    | Aya,Suha,Tamara |
| 2.1.4 Chapter 2 Approval                | 0 days         | Mon 22/12/25        | Mon 22/12/25        | 16,17,18 | M.Obaid         |
| <b>2.2 System Analysis</b>              | <b>59 days</b> | <b>Mon 20/10/25</b> | <b>Thu 08/01/26</b> |          |                 |
| 2.2.1 Requirements analysis             | 10 days        | Tue 23/12/25        | Mon 05/01/26        | 19       | Aya,Suha,Tamara |
| 2.2.2 Hardware and software analysis    | 15 days        | Mon 20/10/25        | Fri 07/11/25        |          | Aya,Suha,Tamara |
| 2.2.3 System Architecture               | 3 days         | Tue 06/01/26        | Thu 08/01/26        | 22,21    | Aya,Suha,Tamara |

Figure 5.2 :B.Project Management Schedule

|                                         |                 |                     |                     |          |                 |
|-----------------------------------------|-----------------|---------------------|---------------------|----------|-----------------|
| <b>3 Executing Phase</b>                | <b>108 days</b> | <b>Fri 09/01/26</b> | <b>Tue 09/06/26</b> |          |                 |
| <b>3.1 System Design</b>                | <b>7 days</b>   | <b>Fri 09/01/26</b> | <b>Mon 19/01/26</b> |          |                 |
| 3.1.1 System Components                 | 7 days          | Fri 09/01/26        | Mon 19/01/26        | 23       | Tamara,Aya,Suha |
| <b>3.2 System Architecture</b>          | <b>8 days</b>   | <b>Tue 20/01/26</b> | <b>Thu 29/01/26</b> |          |                 |
| 3.2.1 Block Diagram                     | 2 days          | Tue 20/01/26        | Wed 21/01/26        | 26       | Aya,Suha,Tamara |
| 3.2.2 Flowcharts                        | 2 days          | Tue 20/01/26        | Wed 21/01/26        | 26       | Tamara,Aya,Suha |
| 3.2.3 Use Case Diagram                  | 2 days          | Thu 22/01/26        | Fri 23/01/26        | 26,28,29 | Aya             |
| 3.2.4 Sequence Diagram                  | 2 days          | Mon 26/01/26        | Tue 27/01/26        | 30       | Suha            |
| 3.2.5 Activity Diagram                  | 2 days          | Wed 28/01/26        | Thu 29/01/26        | 31       | Tamara          |
| 3.2.6 Chapter 3 Approval                | 0 days          | Tue 27/01/26        | Tue 27/01/26        | 26,30,31 | M.Obaid         |
| <b>3.3 Mobile App Development</b>       | <b>95 days</b>  | <b>Wed 28/01/26</b> | <b>Tue 09/06/26</b> |          |                 |
| 3.3.1 User Interface Design             | 5 days          | Wed 28/01/26        | Tue 03/02/26        | 33       | Aya,Suha,Tamara |
| 3.3.2 Software and AI Model Development | 90 days         | Wed 04/02/26        | Tue 09/06/26        | 35       | Aya,Suha,Tamara |
| 3.3.3 Prototype Implementation          | 90 days         | Wed 04/02/26        | Tue 09/06/26        | 33,35    | Aya,Suha,Tamara |

Figure 5.3 :C.Project Management Schedule

|                                     |                 |                     |                     |    |                         |
|-------------------------------------|-----------------|---------------------|---------------------|----|-------------------------|
| <b>4 Monitoring and Controlling</b> | <b>187 days</b> | <b>Mon 20/10/25</b> | <b>Tue 07/07/26</b> |    |                         |
| 4.1 Task Scheduling                 | 7 days          | Fri 09/01/26        | Mon 19/01/26        | 20 | Aya,Suha,Tamara         |
| 4.2 Regular Meetings                | 1 mon           | Mon 20/10/25        | Fri 14/11/25        |    | Aya,M.Obaid,Suha,Tamara |
| 4.3 Documentation Updates           | 1 mon           | Wed 10/06/26        | Tue 07/07/26        | 24 | Aya,Suha,Tamara         |
| 4.4 Quality Assurance               | 1 mon           | Wed 10/06/26        | Tue 07/07/26        | 24 | Aya,Suha,Tamara         |
| <b>5 Closing</b>                    | <b>11 days</b>  | <b>Wed 10/06/26</b> | <b>Wed 24/06/26</b> |    |                         |
| 5.1 Final Documentation             | 6 days          | Wed 10/06/26        | Wed 17/06/26        | 37 | Aya,Suha,Tamara         |
| 5.2 Project Presentation            | 3 days          | Thu 18/06/26        | Mon 22/06/26        | 44 | Aya,Suha,Tamara         |
| 5.3 Supervisor Feedback             | 2 days          | Tue 23/06/26        | Wed 24/06/26        | 45 | M.Obaid                 |
| 5.4 Project Approval                | 0 days          | Wed 24/06/26        | Wed 24/06/26        | 46 | M.Obaid                 |

Figure 5.4 :D.Project Management Schedule

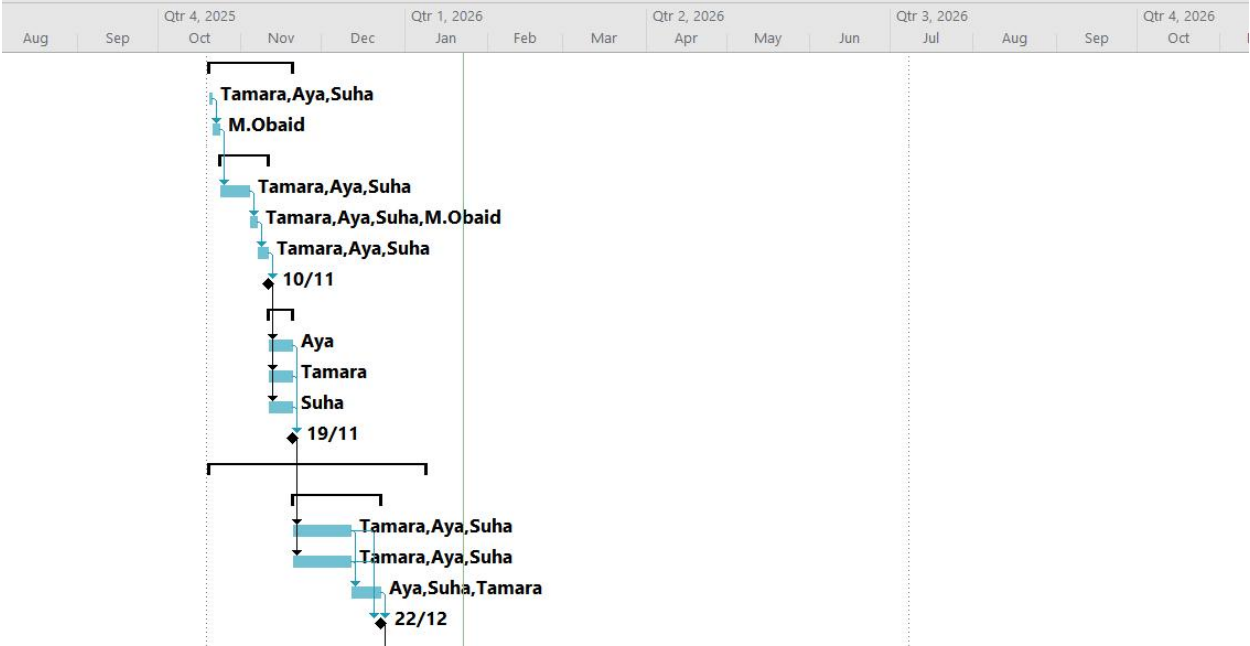


Figure 5.5 :E.Project Management Schedule

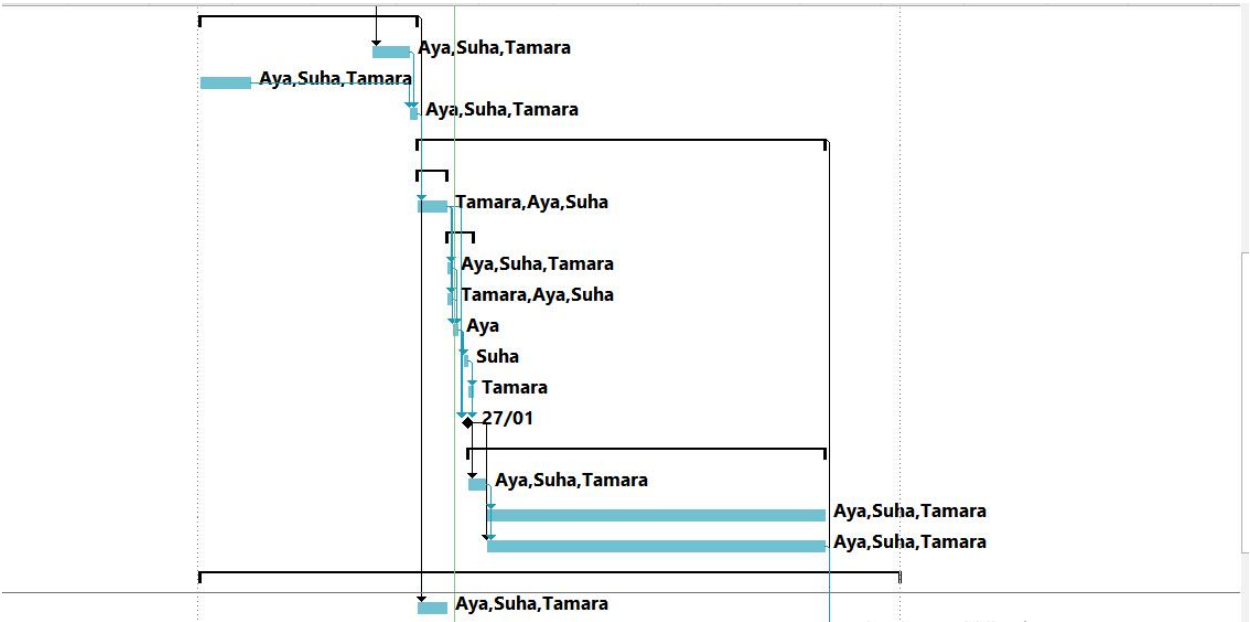


Figure 5.6 :F.Project Management Schedule

## 5.2 Resources and Cost Management

This section discusses the estimated costs of developing and implementing the proposed system for analyzing and monitoring the soil system during the design and development phase. In the estimate, some of the necessary resources to be used to develop the system are discussed. The necessities include the hardware resources that are necessary to design the proposed system of a mobile intelligent vehicle, sensors, communication devices, power supply, and the necessary software resources used to design the mobile app.

The cost of each component is worked out independently of the total cost to give a clear picture of the budget that will be required to complete the project. This estimate is intended to guarantee the economic viability of the system with appropriate performance and reliability. The following table illustrates the allocation of resources that will be required to complete the project successfully.

*Table 3 :Development costs*

| Item                                 | Unit Cost |
|--------------------------------------|-----------|
| Sensors (DHT22, DS18B20, and HW-080) | ₦75       |
| Battery and Base                     | ₦ 45      |
| GPS Module NEO-6M                    | ₦ 25      |
| ESP32                                | ₦ 45      |
| Breadboard and Wire                  | ₦ 30      |
| RC Excavator Vehicle                 | ₦ 150     |
| Total senior project estimate        | ₦ 370     |

## 5.3 Software Used

### 1) Jira

Jira Software: This project management platform, through a workplace organized into boards, keeps you organized. It helps groups work together by letting you assign work, set deadlines, and track progress in real-time while providing lots of issue-tracking features[\[20\]](#). It helps you to summarize your project with milestones, calendars, and summary views. Jira was used in our project to decompose the development process into further shred-down tasks to ensure deadlines were being met, making collaboration and time management very efficient.

### 2) MS Project

MS Project is a Microsoft project management application used for Gantt charts, schedules, and resource management. The tool enables us to assign tasks to team members and manage progress on the project[\[21\]](#). We use MS Project to make and manage detailed project plans with MS Project's powerful scheduling engine. MS Project allows us to make task dependencies and resource assignments. It allows us to maximize resource utilization and for tracking project progress.

### 3) MS Word

Microsoft Word is a powerful word processing application widely used for document creation, formatting, citation management, and collaboration[\[22\]](#). We use it in Smart soil analysis for creating, editing, and formatting rich text documents using templates, styles, tables, images and more.

### 4) Figma

Figma is a cloud-based interface design and prototyping platform that empowers designers to create in real time, offering vector editing, wireframing, interactive

prototypes, and whiteboarding directly in the browser or desktop apps[\[23\]](#). In our project, we use Figma to develop interactive prototypes and create UI/UX mock-ups.

## 5) Arduino IDE

The programming of the ESP32 micro-controller was done through Arduino IDE. It was used to obtain data from the sensors, analyze soil data, connect with the GPS module, and send wireless information to the application[\[24\]](#).

## 6) Flutter

The mobile application for the system was created using Flutter. With the help of the framework, the modern interface for displaying information about the soil measurements, crop suggestions, and location was created[\[25\]](#).

## 7) Firebase

The Cloud Back-end Platform that was used to store the sensor data, user data, and the historical data was Firebase. Firebase also provided real-time synchronization between the mobile app and the database[\[26\]](#).

## 8) Edge Impulse Studio

Edge Impulse Studio was employed to design, train, and test the machine learning model that recommended crops. The Crop Recommendation *Data set* was uploaded into Edge Impulse. Data pre-processing, feature scaling, model training, validation, and testing were done within Edge Impulse. The final trained model had an accuracy of 97.9% after validation[\[27\]](#).

## 9) Android Studio

Development of the mobile application of the proposed system was done using Android Studio as the main development tool. The tool helped in providing means to design the user interface, write application code in Flutter, debug the application, test the application, and integrate with Firebase services[28].

## 10) MATLAB

MATLAB was used to preprocess and prepare the crop recommendation dataset before it was uploaded to Edge Impulse Studio for model training, validation, testing, and deployment[29].

## 5.4 Lessons Learned

During the execution of this graduation project, several positive and negative practices were documented, contributing to a comprehensive learning experience for the team.

Advance planning and organized task distribution among members helped improve workflow and ensured that the phases of the project were completed on time. Collaboration among members continuously helped resolve technical issues, especially during hardware and software integration. Moreover, phased testing and repeated trials increased system stability and accuracy of results.

On the other hand, there were also some challenges and negative practices that the delivery of some electronic components was late; this had a consequence of delaying the final test phases. Additionally, solving compatibility problems between sensors and *micro-controllers* also took more time than necessary for testing and calibration. Besides, not always having a stable network connection made it difficult to test real-time data transmission.

These lessons learned, in general, reinforced the technical and managerial skills of the team and improved the capability for solving real-life engineering problems, thus providing a good foundation for future projects.

## 6 CHAPTER 6: IMPACT OF THE ENGINEERING SOLUTION

The potential impact that this proposed system of intelligent soil analysis and monitoring system development can have on engineering and agriculture, especially with regards to enhancing soil analysis efficiency, decision-making, and encouragement of smart technology usage in field applications, is what is highlighted in this chapter. The areas where this system is also expected to have a major impact, with regards to economic, social, and environmental factors, are also discussed in this chapter, together with the ethics of intelligent monitoring systems and IoT.

### 6.1 Economical, Societal and Global

#### 1) Economical:

The suggested smart soil analysis and monitoring system can provide multiple benefits related to economy, as it reduces the cost and effort that go into soil data collection and analysis. Even though the system is currently created in a form of a prototype for educational purposes, it is essential to evaluate the economic feasibility of the system as well.

##### 1. Prototype Production Cost and Design

The prototype was designed using low-cost, commercially available components, such as:

- ESP32 microcontroller
- Sensors (DHT22, DS18B20, and HW-080)
- NEO-6M GPS module
- Remote-controlled excavator vehicle

The cost of creating the prototype will amount to around 370<sup>TL</sup>, depending on the chosen electronic components and sensors. Such prices make it possible to develop the system for educational projects and agricultural field tests.

## 2. Cost of Mass Production

For large-scale production, the overall system cost can be reduced by:

Buying electronic components in larger quantities

- Using standard hardware platforms
- Simplifying the process of assembling and integrating parts
- Automated maintenance and monitoring, which will lower the related costs

Finally, the end product will be economically efficient compared to the traditional approach to soil testing and analysis in laboratories.

## 3. Cost Saving Benefits

The system contributes to cost reduction by:

- Reducing the need for expensive laboratory tests.
- Less time and effort are needed for soil sampling manually.
- Real-time soil monitoring and faster decision making.
- Better agricultural resource management and monitoring.

## 4. Tax Incentives and Environmental Benefits

The system uses low-power electronic components such as ESP32 micro-controller, thus lowering energy consumption. Moreover, the use of real-time soil monitoring can lead to increased efficiency of agriculture and reduce unnecessary water and fertilizer consumption.

## 2) Societal:

The proposed smart system for analyzing and monitoring the ground plays a part in bringing about considerable benefits to society, mainly in the fields of agriculture and engineering. This system contributes to enhancing the productivity of farm work by allowing access to pertinent information regarding the ground conditions, thus improving the quality of construction work performed.

- Life Impact

The system enables the user to reduce time and effort needed for soil testing, as data can be directly obtained for analysis at the work site. Additionally, the usage of a smart vehicle makes the testing of the soil safer for workers as they might be exposed to hazardous areas.

- Addressed Needs

The system will help to solve the increasing problem of dependable real-time soil information, especially for rural or remote areas with limited laboratory facilities or technical expertise. It will also improve decision-making in the agricultural sector as well as agricultural engineering projects.

- The shift towards smart on-demand monitoring

The proposed solution promotes a change from traditional and human-dependent approaches to soil testing to smart technology, which will utilize the Internet of Things and artificial intelligence technologies, allowing users to give themselves more opportunities for self-monitoring and making independent decisions quickly.

### 3) Global:

At the global level, the smart soil analysis and monitoring system can help with improved food security, land sustainability, and infrastructure construction. The system can be applicable in many different geographic locations worldwide, including developing countries.

The integrated technologies of IoT and AI within the system create potential for monitoring soils under different environmental and climatic conditions, contributing to optimal agricultural production and resource conservation across the world. In addition, the system of soil moisture monitoring technologies encourages the adoption of smart technologies worldwide, facilitating a closing gap between technologically developing and developed countries.

## 6.2 Environmental and Ethical

The basic purpose of the proposed system of intelligent analysis and monitoring of soil is to be used in agricultural fields; however, its development and usage also raise various environmental and ethical issues as well.

From an environmental point of view, this system has little detrimental effects on the environment due to the small and low-power electronic components it uses, such as micro-controllers and batteries. It does not require heavy and costly equipment, thus minimizing the usage of natural resources such as water.

The system also helps in reducing environmental impacts, as it minimizes the use of traditional soil tests, which would require moving the soil, consuming energy in the laboratories, or spending resources on transportation. Additionally, the system reduces the emissions associated with moving zones and relocating the system.

Besides, the system comes along with moderate energy consumption, and it is not meant to operate even during idle times, thereby preventing energy wastage. All these contribute to making it an effective means of supporting energy-efficient, environmentally friendly, and sustainable engineering practices.

Ethically, the system aims at accumulating data on soil and geography only for analysis purposes while maintaining privacy levels of the users. It is also important to ensure that data that has been accumulated is kept safe and cannot be accessed by unauthorized personnel, especially when conducting important agricultural or engineering projects.

### 6.3 Other Issues

In addition to various environmental, economic, and social issues, various organizational and operational aspects also have to be taken into consideration for the success and reliability of the system.

One major issue with this system is the accuracy of the data obtained, since the system's functionality relies on accurate sensors and properly obtained data. Any malfunction in the acquired data can cause consequent faulty decisions, thereby requiring periodic calibration of these sensors used in the smart vehicle.

The dependability of models of AI employed in the analysis of the soil composition will need to be ensured through training the models on data that is representative of different types of soils and varied environmental conditions.

Last but not least, the ease of use of the system and the simplicity of the user interface are very important, which would help farmers and agricultural engineers with different technical backgrounds utilize the system properly, thus encouraging people's acceptance and adoption of the system.

## 7 CHAPTER 7: CONCLUSIONS AND RECOMMENDATIONS

The present chapter provides a comprehensive summary of the main achievements and accomplishments recorded by the Smart Soil Analysis and Monitoring System project, showing the extent to which the objectives were met, focusing on the results attained within the phases of design and development of the system. Furthermore, this chapter incorporates the skills attained by the team throughout the implementation process, showing aspects such as technology, analysis, and teamwork.

### 7.1 Project Achievements and Conclusions

The Smart Soil Analysis and Monitoring System project succeeded in integrating embedded systems, wireless communications technology, artificial intelligence, and smart agriculture concepts into one single concept.

In this regard, the new system incorporates a remotely controlled excavator machine, which analyzes the soil using the Sensors (DHT22, DS18B20, and HW-080). The sensors measure soil moisture and soil temperature value and air moi moisture, air temperature. Information obtained from sensors will be analyzed using ESP32 micro-controllers and transmitted wireless using Wi-Fi and Bluetooth technologies.

Also, a GPS sensor was used to capture the geographical location corresponding to the soil reading. First, the preprocessing of the crop recommendation dataset was performed using MATLAB. It was then imported into Edge Impulse Classification Learner for feature scaling, model training, validation, testing, and deployment.

The project was an example of how hardware systems, software applications, AI, and mobile devices could be merged to create a practical tool for smart agriculture applications.

In summary, the developed system represents a promising and affordable way to make soil analysis easier for farmers and agricultural specialists and improve agricultural decision-making.

## 7.2 Skills and Experience Gained

In the process of designing and implementing the current project, the team has acquired many technical and practical skills connected with embedded systems, smart agriculture technologies, artificial intelligence, and collaboration.

This project has allowed acquiring some practical skills for dealing with ESP32 micro-controllers, wireless technology, GPS systems. Also, the team learned how to develop an intelligent system using hardware such as the Sensors (DHT22, DS18B20, and HW-080), and mobile application.

However, it is essential to note that the use of artificial intelligence and machine learning technology was also employed in the project. MATLAB was used for pre-processing of the dataset on crop recommendation and then imported to Edge Impulse Classification Learner to train, validate, test, and deploy the AI classifiers to achieve better prediction results and recommend appropriate crops.

In addition, this project helped us learn about IoT applications, mobile app interfaces, real-time data processing, and wireless technologies.

Professionally, it is important to mention that teamwork, project management skills, communication skills, technical writing, and problem-solving techniques were developed by the team members throughout the project period.

### 7.3 Recommendations for Future Work

A number of potential improvements for the future could include:

One of the future upgrades that could be developed for the current design is creating an autonomous excavating vehicle which can move around and sample soils automatically without constant manual operation.

The system could also be supplemented with various environmental sensors which measure parameters such as the salinity of soils, air quality, and weather conditions in order to conduct more comprehensive agricultural analysis.

An improvement could be achieved through the use of larger and more diverse datasets used in training of the AI algorithm to improve prediction accuracy and increase crop recommendation quality.

Using cloud computing and IoT capabilities could be applied to facilitate large-scale agricultural monitoring as well as access to the data gathered from multiple agricultural sites in real time.

Improvements could be made by designing the mobile application to visualize gathered data, conduct historical data comparisons and utilize live GPS tracking with notifications capabilities.

Other improvements might be focused on achieving higher energy efficiency and wireless communication reliability.

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## APPENDICES

**A p p e n d i x   A :**

**A p p e n d i x   B :**

**A p p e n d i x   C :**